COMPUTER VISION - ASSIGNMENT 2

Develop and Evaluate an Image Enhancement and Segmentation System for Industrial Metal Surface Inspection

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Course code: ITA0505 -Computer vision

| CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images.

0. Simulated image set, calibration and evaluation method

0.1 The simulated image set

A generator (common.py, seeded for reproducibility) builds 512 x 512, 8-bit grayscale images of brushed steel: horizontally streaked texture (std about 4.2 grey levels), sensor noise (sigma 1.5), and harmless **rolling marks** (long faint horizontal lines, 3-5.5 grey levels) that the system must not report. Four capture conditions model the factory: **normal**, **dim** (about 0.48x brightness), **uneven** (linear gradient plus vignetting) and **reflective** (a bright saturating highlight band and a dark reflection band). Defects are added in absolute grey levels after the illumination is applied, so a "low-contrast" defect really differs from healthy metal by 6-12 grey levels as the brief specifies.

Item	Value
Images	66 total = 60 main images (15 per condition; 8 defect-free) + 6 sheet-boundary images (dark background strip with a sharp edge)
Labelled defects	107 instances in the 60 main images: scratch 29, pit 26, stain 20, crack 18, scale mark 14, pinhole 9 (counts include the boundary images for the generator summary; tasks use the main set unless stated)
Low-contrast defects	About 40 % of pits, scratches, cracks and all stains have peak contrast of 6-12 grey levels
Distractors (not defects)	Rolling marks (every image), dust specks 1-2 px bright (45 % of images), reflections, sheet boundary
Ground truth	Per-image instance label map (type, polarity, low-contrast flag) saved as .npy, plus meta.json - original images are never modified
Scale / calibration	512 px = 80 mm field of view -> **0.156 mm per pixel**: a 0.3 mm scratch is about 2 px wide, a 0.5 mm pit about 3 px diameter

0.2 Evaluation measures used throughout

- **Absolute contrast** = |mean(defect) - mean(healthy ring)| in grey levels, where the ring is the 7-19 px band around the defect (displayed contrast).
- **CNR (contrast-to-noise ratio)** = absolute contrast / standard deviation of the ring (detectability; a point operation that scales everything equally cannot change CNR).
- **Candidate detector (fixed rule for all enhancement tasks):** deviation from a Gaussian-blurred background (sigma 10), threshold 4 x robust sigma (MAD), components of at least 6 px. A **false alarm** is a component not touching any labelled defect (dust, rolling marks and grain count as false alarms); **recall** is the fraction of labelled defects touched by a component.
- **Pixel metrics** (Tasks 8, 10): precision, recall, IoU = TP/(TP+FP+FN) and Dice = 2TP/(2TP+FP+FN) against the reference masks. **Edge metrics** (Task 7): precision/recall with a 2 px tolerance.
- **Timing:** median of repeated runs on a 512 x 512 image, single CPU core of the sandbox (the edge PC has 4 cores, so parallel tiling can only make this faster).

0.3 Representative images

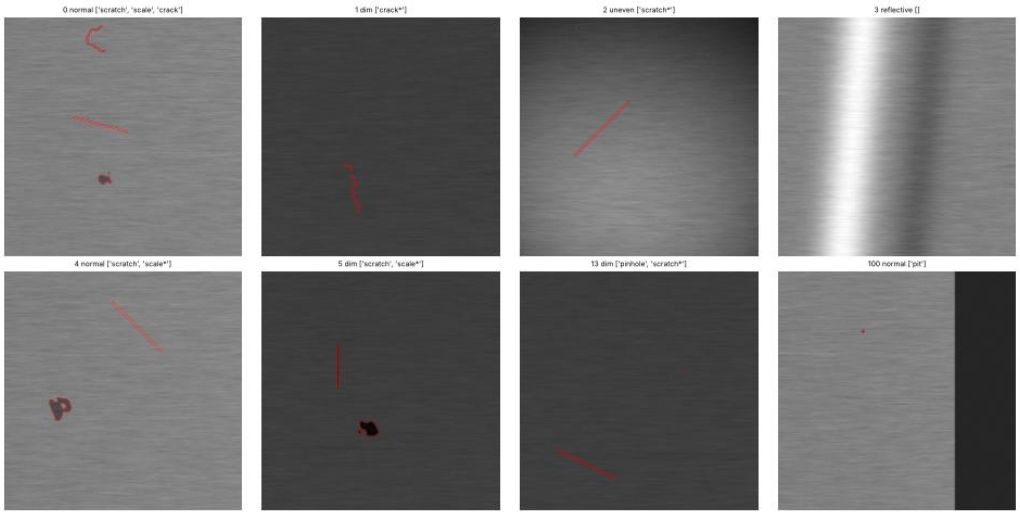


Figure 0.1 - Examples from the simulated set (red = reference defect masks). Top row: normal, dim, uneven, reflective-sheet examples; the right-most image has a sheet boundary.

0.4 Limitations of the simulation (apply to every task)

- Texture, noise and defect shapes come from simple statistical models; real mill finish has coarser, correlated structure, specular glints and colour/oxide variation that this simulation does not reproduce.
- Defect positions are known exactly, but real manual labelling has its own uncertainty (about +/-1 px at 0.3 mm scratch width).
- The simulated field of view is 80 x 80 mm. The full 600 x 400 mm sheet needs about 38 such windows at the same resolution; Section 10.5 shows the effect on the 250 ms budget.
- Sample sizes are small (tens of images, 5-30 instances per defect type), so differences of a few percentage points are not statistically meaningful; large differences and mechanisms are.

Task 1 - Spatial-domain enhancement I: grey-level transformations [4 marks]

Prediction (formed from theory before running the code): Linear stretching and piecewise-linear mapping will multiply displayed contrast by several times but cannot raise CNR (they scale texture and defect together), and any window that clips defect pixels will lower it. Gamma/log will help only the dark (dim) images. The piecewise map should give the best control because it can compress strong scale marks while expanding the faint 6-12 level range.

1.1 Transformations implemented (lookup tables, cv2.LUT)

Method	Equation / lookup table	Parameters and how chosen
A Linear stretch	s = 255 (r - r_min)/(r_max - r_min), clipped to [0,255]	r_min, r_max = 0.1 and 99.9 percentiles of each image (mean 59 and 161, range 0-106 and 71-255). Chosen from a test of 4 percentile pairs (Table 1.3): 0.1/99.9 keeps 97 % of the CNR with 0.37 % clipped pixels. Slope = 255/(r_max - r_min), about 2.5.
B Gamma (and log)	s = 255 (r/255)^g ; log: s = 255 ln(1+r)/ln(256)	g chosen per image so the median maps to 128: g = ln(0.5)/ln(median/255), limited to 0.4-2.5; measured g = 0.85 on average (0.49 dim ... 1.09 bright). Input range 0-255, output range 0-255; slope < 1 for bright, > 1 for dark.
C Piecewise linear	three segments: [0,r1] -> [0,40]; [r1,r2] -> [40,215]; [r2,255] -> [215,255]	median m and noise sigma s measured from the image (MAD of the high-pass residual, mean 3.65). r1 = m - 4s (mean 93.5), r2 = m + 4s (mean 122.7). Slopes: 0.43 (strong dark marks, compressed), about 6.0 (healthy metal + faint defects, expanded), 0.30 (highlights, compressed).

Original images are never modified; each transformed result is a new array (saved in the results folder of the code archive). The test used **24 images** with scratches, cracks or pits (36 defect instances, 12 of them low-contrast) plus normal brushed texture in every image.

1.2 Measured comparison

Method	Mean contrast (grey levels)	Contrast, low-contrast defects	CNR (all)	CNR (low-contrast)	Image std	Clipped px (0/255)	Time (ms)
Original	16.5	6.5	3.63	1.52	22.6	0.16 %	-
A Linear stretch (0.1-99.9 %)	44.8	24.3	3.52	1.47	46.0	0.37 %	0.09
A2 Linear p1-p99 (unsuitable)	53.2	37.5	2.37	1.35	61.2	2.48 %	0.08
B Gamma (median to 128)	17.1	6.6	3.86	1.54	23.2	0.16 %	0.08
B2 Log	8.1	3.2	4.23	1.56	8.6	0.16 %	0.08
C Piecewise linear	51.1	28.6	2.61	1.41	55.1	0.16 %	0.08

Table 1.1 - Contrast, histogram spread, clipping and time over 24 images / 36 defects.

Condition	Original: contrast / CNR	A Linear	B Gamma	C Piecewise
normal	17.7 / 4.0	66.9 / 3.5	17.7 / 4.0	60.0 / 2.7
dim	17.1 / 7.5	90.0 / 7.4	19.7 / 8.7	80.5 / 3.8
uneven	13.9 / 2.9	30.8 / 2.9	13.9 / 3.1	23.7 / 1.6
reflective	17.3 / 2.3	26.3 / 2.3	17.7 / 2.3	52.3 / 2.7

Table 1.2 - Contrast (grey levels) / CNR by capture condition.

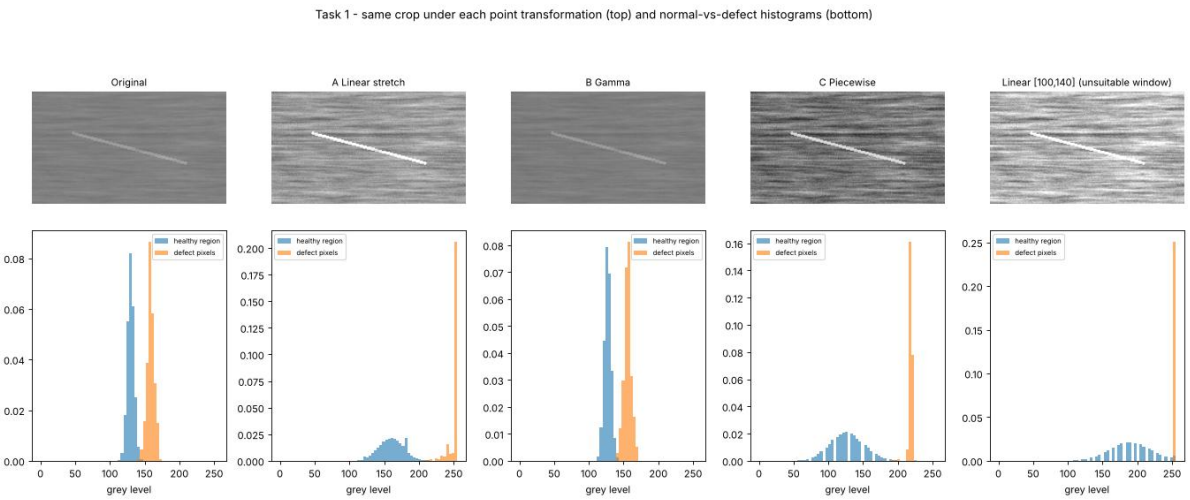


Figure 1.1 - The same defective crop under each point transformation (top) and the histograms of healthy pixels versus defect pixels (bottom). The right-hand column is an unsuitable parameter choice.

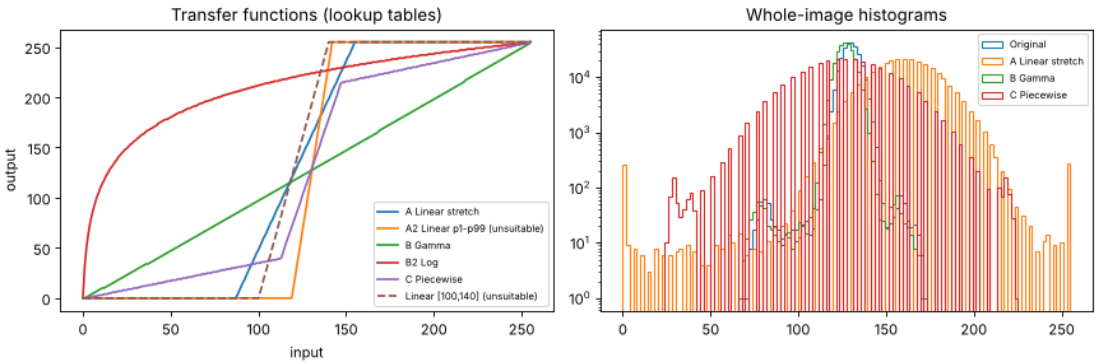


Figure 1.2 - Transfer functions (lookup tables) and whole-image histograms before and after transformation.

1.3 Parameter selection and an unsuitable parameter

Linear stretch limits. Four percentile pairs were tested on the same 24 images: 1-99 % gave CNR 2.37 with 2.5 % of pixels clipped, 0.1-99.9 % gave 3.52 (0.37 % clipped), 0.01-99.99 % gave 3.62 and the raw min-max range gave 3.63 (but only 15 grey levels of mean contrast on low-contrast defects against 24.3 for 0.1-99.9 %). The 0.1/99.9 pair is therefore the best compromise.

Faint-defect example, parameter choice	Contrast (grey levels)	CNR	Clipped px
Original (no transform)	30.7	6.79	0 %
A Linear 0.1-99.9 % (selected)	98.4	5.82	0.21 %
A Linear 1-99 % (too tight)	153.8	3.18	2.57 %
A Linear window [100,140] (far too narrow)	76.2	2.65	2.74 %
B Gamma 0.3 (too low)	13.9	6.13	0 %
B Gamma (selected)	30.7	6.80	0 %
C Piecewise (selected)	101.3	4.41	0 %

Table 1.3 - One strong scratch under good and unsuitable parameters. A too-tight window drives the defect pixels to the clip value: displayed contrast rises (154 grey levels) while CNR falls from 6.8 to 3.2, so the scratch becomes less distinct from the texture; the window [100,140] saturates 2.7 % of the image and drops CNR to 2.6. Gamma 0.3 compresses the mid/bright range so the absolute contrast falls to under half.

1.4 How the mappings change visible intensity differences

In the histograms (Fig. 1.1) the healthy-pixel distribution of this brushed metal is a narrow peak about 25-40 grey levels wide, and the faint defect sits only 6-12 levels from its centre, so on the display it is nearly invisible. Linear stretching multiplies every difference by the same slope (about 2.5), so the peaks widen but their overlap, and therefore CNR, is unchanged; the measured mean contrast of low-

contrast defects rises from 6.5 to 24.3 grey levels and CNR stays at 1.47 versus 1.52. The piecewise map concentrates its steep slope (about 6) on m +/- 4 s, so low-contrast defects reach 28.6 grey levels while scale marks (far below r1) are compressed rather than driven into black. Gamma with g near 0.85 is almost the identity on bright sheets, but on the dim sheets (g about 0.5) it lifts CNR from 7.5 to 8.7 because it reallocates the 8-bit levels toward the dark region where quantisation had been coarse.

1.5 Comparison for low-contrast defects and reflective regions

- **Low-contrast defects:** all three give CNR within 0.1 of the original (1.4-1.6): **point transformations improve what the operator can see, not what is statistically detectable.** Display contrast of faint defects: linear 24.3, piecewise 28.6, gamma 6.6, log 3.2 (log compresses the bright range and reduces contrast).
- **Reflective sheets:** the highlight pushes the 99.9th percentile to 255, which wastes the linear stretch (mean contrast only 26 versus 90 on dim sheets, CNR unchanged at 2.3). Piecewise is the only method that raises contrast there (52 grey levels, CNR 2.7) because its break points follow the **median** and the noise sigma, not the extreme percentiles, so the highlight does not set the slope.
- **Strong scale marks:** piecewise lowers the CNR of strong defects (overall CNR 2.6 versus 3.6) because it deliberately compresses them - acceptable, since they are already obvious.

1.6 Recommendation

> **Use the piecewise-linear transformation (C), controlled by the measured median and noise sigma, as the display/inspection mapping, optionally preceded by gamma on dim sheets.** It gives the largest display contrast gain for faint defects (x4.4), does not clip the healthy range, and is robust to highlights. Processing cost is negligible (0.09 ms with a lookup table).

When not to use it: (i) not as the input to a threshold calibrated on raw images - the mapping changes the grey-level scale, so thresholds must be set in the mapped domain; (ii) not when the median is itself biased by a large defect or a very wide highlight (more than about 30 % of the image), where r1 and r2 move away from healthy metal; (iii) never as evidence - because CNR is unchanged, it cannot prove a defect exists.

> **Outcome vs prediction:** Confirmed: CNR is unchanged or slightly lower for A and C while contrast is multiplied by 3-4; gamma helps only dim sheets; clipping windows hurt. Not confirmed: I expected the piecewise map to stay close to the original CNR - it is lower overall (2.61 vs 3.63) because it intentionally compresses strong defects.

Task 2 - Spatial-domain enhancement II: histogram processing [5 marks]

> **Prediction (formed from theory before running the code):** Global histogram equalisation (HE) will improve contrast in a uniformly lit image but, with uneven lighting or a bright highlight, the histogram is dominated by the illumination so local defects lose contrast; it will amplify noise strongly. Unlimited AHE will amplify grain and create false alarms. CLAHE with a small clip limit should be the best compromise.

2.1 Procedure

Implemented with cv2.equalizeHist (A), cv2.createCLAHE with an effectively unlimited clip limit (clipLimit = 1000, 8 x 8 tiles) as adaptive HE (B - note OpenCV interpolates between tiles, so this is "AHE with bilinear tile blending"), and cv2.createCLAHE(clipLimit, tileGridSize) (C). **40 images** (10 per condition: normal, dim, uneven, reflective-highlight; 70 defects) were used. Local contrast is measured between each labelled defect and its surrounding ring; false alarms, noise amplification (sigma of healthy region after / before), clipped pixels and time are recorded.

2.2 Comparison

Method	Local contrast (grey lv)	CNR (all)	CNR (low-contrast)	Recall	False alarms / image	Noise amplification	Clipped px	Time (ms)
Original	21.3	5.26	3.50	66%	0.8	x1.00	0.12 %	0.00
A Global HE	74.4	2.42	1.27	44%	145.9	x11.80	0.50 %	0.17
B AHE (8x8, no clip)	92.2	2.16	1.68	9%	26.2	x11.46	0.39 %	0.91
C CLAHE (clip 2, 8x8)	36.2	3.48	2.40	53%	0.3	x1.81	0.11 %	0.91

Table 2.1 - Comparison on 40 images / 70 defects (fixed candidate-detector rule, threshold 4 sigma).

Condition	Original (contrast / CNR)	Global HE	AHE	CLAHE 2 / 8x8
normal	24.9 / 5.8	100.3 / 1.5	102.1 / 1.5	40.6 / 3.4
dim	25.5 / 11.1	126.6 / 1.9	127.1 / 1.9	39.1 / 5.6
uneven	20.2 / 4.7	53.6 / 4.1	90.2 / 2.3	35.4 / 3.6
reflective	17.6 / 2.1	44.8 / 2.1	68.5 / 2.6	32.7 / 2.3

Table 2.2 - Defect-to-background contrast (grey levels) / CNR per capture condition.

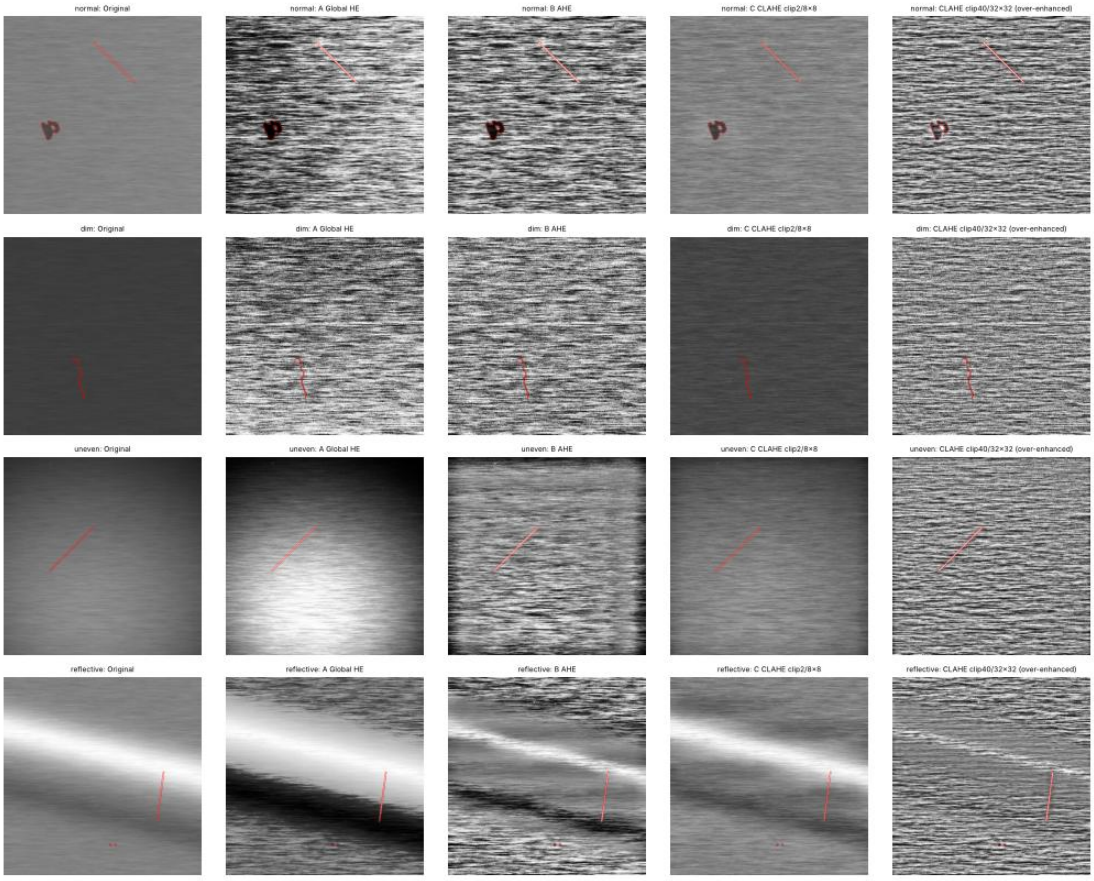


Figure 2.1 - Before/after for the four conditions (rows) and methods (columns). The last column (clip 40, 32 x 32 tiles) is deliberately over-enhanced: grain becomes artificial-looking contrast. Red outline = reference defect.

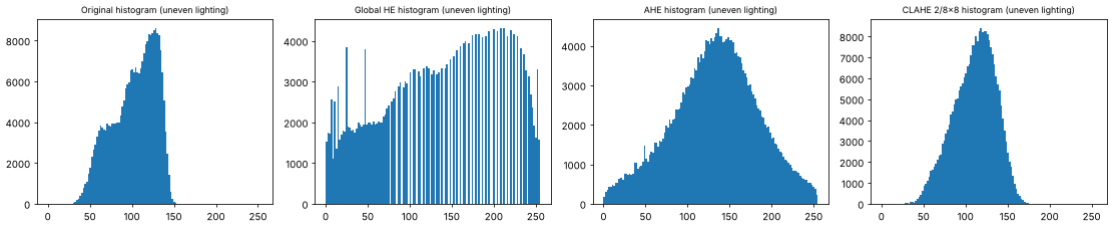


Figure 2.2 - Histograms for an unevenly lit sheet: the illumination gradient controls the global-equalised histogram.

2.3 Why global equalisation can hide a local defect

Global HE builds one mapping from the whole-image histogram. On an unevenly lit sheet the histogram is a broad sum of the illumination gradient, so the cumulative distribution is nearly linear over the whole range and the mapping stretches **illumination**, not texture. Healthy-region noise grows by x11.8 and, on the dim sheets, the 6-12 level scratch is spread into the noise: its CNR falls from 7.5 (original) to 1.9 in the dim images, and in the normal-illumination images from 4.0 to 1.5. Contrast looks higher (74 grey levels versus 21) because both defect and grain are expanded, and the detector then reports **146 false alarms per image**. Where the defect is in a dark corner but the histogram mass is in the bright region, the corner is compressed by the steep part of the cumulative distribution and the defect becomes less distinct while another region of the same image looks better.

2.4 Sensitivity of CLAHE (tile size and clip limit)

Clip limit	Tiles	Local contrast	CNR	CNR low-contrast	Recall	False alarms / img	Noise amp.
1.0	8 x 8	29.8	4.00	2.65	63%	0.45	x1.39
1.0	16 x 16	31.0	4.06	2.77	63%	0.23	x1.43
1.0	32 x 32	30.1	3.33	2.27	57%	0.50	x1.44

Clip limit	Tiles	Local contrast	CNR	CNR low-contrast	Recall	False alarms / img	Noise amp.
1.5	8 x 8	33.1	3.69	2.51	56%	0.25	x1.61
1.5	16 x 16	34.0	3.74	2.67	56%	0.17	x1.63
1.5	32 x 32	30.1	3.33	2.27	57%	0.50	x1.44
2.0	8 x 8	36.2	3.48	2.40	53%	0.33	x1.81
2.0	16 x 16	36.9	3.48	2.50	51%	0.15	x1.80
2.0	32 x 32	36.7	2.89	2.07	53%	0.38	x1.87
4.0	8 x 8	46.6	3.07	2.23	36%	0.10	x2.66
4.0	16 x 16	47.6	2.90	2.23	31%	0.05	x2.63
4.0	32 x 32	48.9	2.68	2.00	37%	0.05	x2.74
10.0	8 x 8	68.4	2.52	1.96	9%	1.13	x5.19
10.0	16 x 16	70.4	2.29	1.87	1%	0.00	x5.11
10.0	32 x 32	67.0	2.12	1.74	0%	0.00	x5.13
40.0	8 x 8	92.1	2.16	1.68	9%	26.18	x11.38
40.0	16 x 16	95.8	1.88	1.54	0%	0.00	x11.37
40.0	32 x 32	89.2	1.59	1.37	0%	0.00	x11.21

Table 2.3 - CLAHE sensitivity (selected rows; full 6 x 4 grid in the code archive). Highlighted row = chosen setting.

Over-enhancement setting: clip 40 with 8 x 8 tiles (effectively unlimited) raises noise x11.4, produces 26 false alarms per image and reduces recall to 9 %; with 32 x 32 tiles at clip 10-40 recall falls to 0-2 % (the tile statistics are dominated by the defect itself, so the defect is equalised away). Small clip limits (1-2) keep noise amplification at x1.4-1.8.

2.5 Selected method and parameters

CLAHE with clip limit 1.5 and a 16 x 16 tile grid (32 x 32 px tiles = 5 mm), 8-bit input/output, used for the operator display image. At this setting local contrast is 34 grey levels (x1.6 the original), noise amplification only x1.6, 0.17 false alarms/image and no extra clipping (0.11 %). It runs in about 1 ms.

Keeping settings consistent across shifts and batches: (1) the parameters are fixed in a version-controlled configuration file, not edited at the console; (2) the image tile grid is defined in millimetres of sheet, not pixels, so a change of camera resolution changes the pixel grid but not the physical scale; (3) a calibration target (grey step wedge and a reference sheet with a known artificial defect) is imaged at the start of each shift, and the system checks that the CLAHE output of the reference defect keeps a CNR within +/-15 % of the stored value; (4) the configuration hash is stored with every result so any change is traceable.

Enhancement is not proof of a defect. Table 2.1 shows that global HE produced 146 false features per image from pure texture and noise: a feature visible only after enhancement may be an artefact of the enhancement. The **original image and the enhanced display are both retained and shown side by side** at review, and reject decisions are made on the measured region (Tasks 8-10) and the original data, never on the enhanced picture alone.

Outcome vs prediction: Confirmed: HE and AHE amplify noise (x11-12), cause large false-alarm counts, and CLAHE at a small clip limit is the safest. Not confirmed: I expected CLAHE to raise detectability; instead the original image has higher CNR (5.26 versus 3.5-4.0 for CLAHE) and recall (66 % versus 51-63 %). Local equalisation is a nonlinear display aid: it increases how visible faint defects are to the eye (contrast x1.6) but slightly reduces statistical detectability. The detection stages in Task 10 therefore work on the unequalised image, and CLAHE is kept in the display path.

Task 3 - Spatial filtering I: smoothing filters [4 marks]

Prediction (formed from theory before running the code): Mean and Gaussian filters reduce Gaussian noise but blur scratches; the median filter removes impulse (salt-and-pepper) noise almost perfectly but is not better than Gaussian for Gaussian noise, and it also deletes small isolated pits and pinholes. Larger kernels remove more noise and more scratch contrast.

3.1 Procedure

20 images (17 scratch/crack instances, 36 dust specks, 3 pinholes available for the isolated-pixel tests) were filtered with mean (cv2.blur), Gaussian (cv2.GaussianBlur, sigma derived from the kernel size) and median (cv2.medianBlur) filters at 3 x 3 and 5 x 5. The stored image is the clean reference. **Noise models:** (a) additive Gaussian noise, zero mean, sigma = 10 grey levels; (b) salt-and-pepper noise, density 2 % (1 % black, 1 % white). Noise quality: PSNR against the clean image. Scratch retention = defect contrast after filtering / contrast in the clean image; a scratch is "visible" if CNR >= 3.

3.2 Results

Filter	PSNR Gaussian noise (dB)	Scratch kept (Gaussian noise)	PSNR salt-and-pepper (dB)	Scratch kept (S&P)	Time (ms)
Noisy input (no filter)	28.1	99%	22.5	97%	-
A Mean 3x3	35.9	82%	31.4	80%	0.04
A Mean 5x5	37.1	60%	34.3	59%	0.06
B Gaussian 3x3	35.7	86%	30.7	84%	0.07
B Gaussian 5x5	37.1	76%	32.9	75%	0.11
C Median 3x3	34.6	79%	41.4	71%	0.05
C Median 5x5	36.4	58%	39.2	61%	0.42

Table 3.1 - Noise reduction (PSNR vs clean image) and scratch contrast retained, 20 images. Highlighted: Gaussian 5x5 and median 3x3 (best for their noise type).

Filter (clean image)	Brushed texture kept	Scratch/crack contrast kept	Bright isolated pixel kept (dust)	Dark speck kept (pinhole, n=3)	Scratch visible (CNR>=3)
No filter	100%	100%	100%	100%	47%
A Mean 3x3	88%	83%	31%	44%	47%
A Mean 5x5	83%	61%	10%	18%	41%
B Gaussian 3x3	89%	87%	45%	56%	47%
B Gaussian 5x5	86%	77%	29%	40%	41%
C Median 3x3	89%	72%	28%	37%	47%
C Median 5x5	84%	61%	3%	7%	47%

Table 3.2 - Effect on the structures of interest in the clean images (fraction of original amplitude retained). With only 3 pinholes the dark-speck column is indicative.

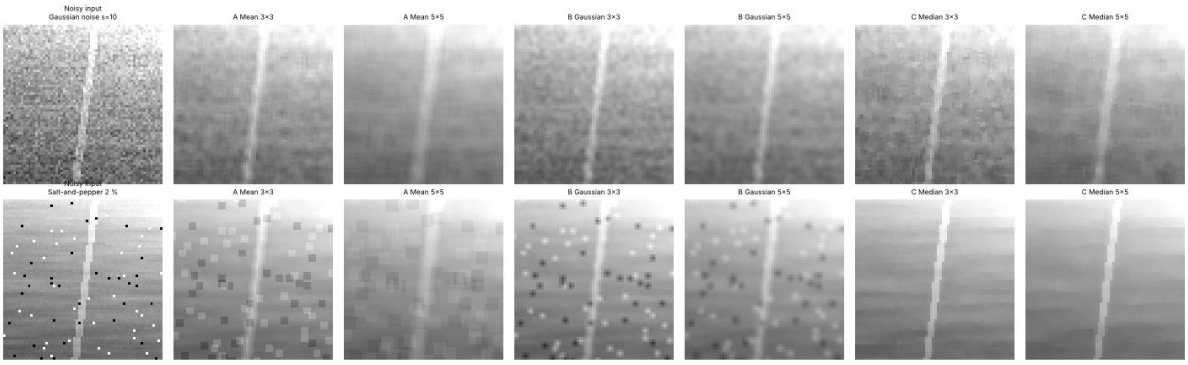


Figure 3.1 - Same 64 x 64 px crop (same scale) containing a scratch: noisy input and all six filter outputs, for Gaussian noise (top) and salt-and-pepper noise (bottom).

3.3 Analysis

- **Gaussian noise:** the 5 x 5 mean and Gaussian filters give the highest PSNR (37.1 dB from 28.1 dB), the median is 0.7-1.5 dB worse. But the 5 x 5 mean keeps only 60 % of the scratch contrast and the 5 x 5 median 58 %, against 76 % for the 5 x 5 Gaussian and 86 % for the 3 x 3 Gaussian.
- **Salt-and-pepper noise:** the median 3 x 3 gives 41.4 dB, 10 dB better than the 3 x 3 Gaussian (30.7 dB) and mean (31.4 dB). A linear filter spreads each +/-127-level impulse into its neighbours (a smudge), whereas the median ranks the pixels and discards an impulse entirely because an outlier never reaches the middle rank. For Gaussian noise every pixel is perturbed slightly, so there are no outliers to reject and averaging (which reduces variance by 1/N) is optimal; the median reduces variance by only about 1/(1.57 N).
- **Isolated structures:** a 5 x 5 median removes 97 % of a bright dust pixel and 93 % of a dark pinhole, and the 5 x 5 mean removes 90 % and 82 %. The 3 x 3 Gaussian retains the most (45 % / 56 %). The same property that removes dust removes a 0.3-0.5 mm real pit (2-3 px).
- **Fine scratches and brushed texture:** a 2 px wide scratch is roughly the width of the kernel, so any kernel of 5 px or more removes 25-40 % of its contrast; texture amplitude falls to 83-89 %. The trade-off is direct: noise PSNR gain per kernel size is bought with scratch contrast.

3.4 Recommendation

Gaussian 3 x 3 (sigma about 0.8) as the default pre-filter for Gaussian-type sensor noise (it keeps 86 % of scratch contrast and 56 % of pinhole depth; 35.7 dB, 0.07 ms), **with a 3 x 3 median only when impulse noise or hot pixels are observed** (a sensor test image with and without lens cap decides). Kernels of 5 x 5 and larger are not used before detection of 0.3 mm scratches. Dust is handled by the area rule in Task 10, not by heavy smoothing.

Outcome vs prediction: Confirmed: median wins for salt-and-pepper (+10 dB), loses ground for Gaussian noise, deletes isolated pits, and larger kernels cost scratch contrast. Only 29 % of scratches are "visible" (CNR >= 3) after any filter against 6 % before in noisy images, showing the benefit of denoising for detection; in the clean set the visible fraction stays at 41-47 %.

Task 4 - Spatial filtering II: sharpening [4 marks]

Prediction (formed from theory before running the code): Laplacian sharpening will give the sharpest edges but also the strongest noise amplification; unsharp masking with unit gain is gentler; high-boost with a gain above 1 gives the strongest scratch depth but overshoots (halos) at strong edges and makes normal grain look like cracks. Sharpening cannot add information and will lower PSNR against a clean reference when noise is present.

4.1 Equations and implementation

Method	Equation	Parameters
A Laplacian	g = f - c * Laplacian(f), kernel [[0,1,0],[1,-4,1],[0,1,0]] (cv2.Laplacian, ksize 1)	c = 1 (c = 0.5 and 2 in the sensitivity study)
B Unsharp masking	mask = f - G_s(f); g = f + k * mask	k = 1, Gaussian sigma s = 1.5 px
C High-boost	g = f + k * (f - G_s(f)) with k > 1 (equivalently g = (A-1) f_blur-removed ... gain above the unsharp case)	k = 3, s = 1.5 px

Tests used **20 images** with thin scratches, cracks, pits and normal texture. A synthetic reference image (brushed texture, one horizontal 0.3 mm scratch of 22 grey levels, and a +40 level block for the edge) is used for the line profile and halo measurement. For PSNR a noise level of sigma 3 was added to each stored image and the sharpened noisy image was compared with the stored image.

4.2 Results

Method	Scratch depth (grey lv)	Max gradient	Scratch FWHM (px)	Edge overshoot	CNR (defects)	Noise amp.	PSNR (dB)	Recall	Time (ms)
Original	22.1	23.3	3.0	3.2 %	4.65	x1.00	38.4	72%	0.00
A Laplacian (c=1)	48.3	71.9	2.0	20.9 %	2.33	x5.36	22.9	43%	0.43
B Unsharp (k=1, s=1.5)	32.4	41.6	3.0	17.0 %	3.97	x1.94	31.9	61%	0.79
C High-boost (k=3, s=1.5)	53.1	78.3	3.0	49.5 %	3.21	x3.81	25.3	57%	0.88

Table 4.1 - Sharpening comparison. Depth, gradient, FWHM and overshoot come from the synthetic profile; CNR, noise amplification, PSNR (sigma-3 noise baseline 38.4 dB) and recall from 20 images.

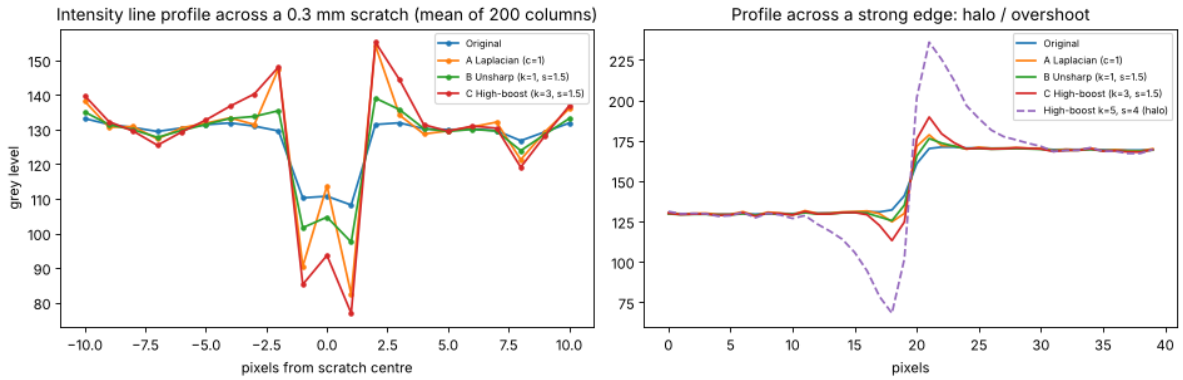


Figure 4.1 - Line profile across a 0.3 mm scratch (left) and across a strong edge (right). All sharpening deepens the scratch; the high-boost curve overshoots the edge plateau by almost 50 %.

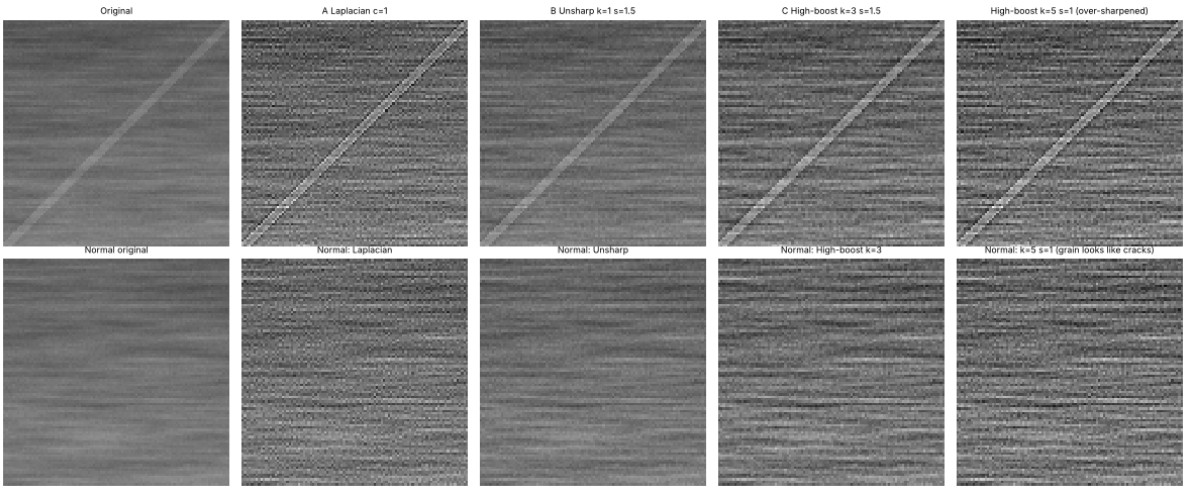


Figure 4.2 - Top: scratch crop under each method and an over-sharpened setting. Bottom: a defect-free region - at k = 5, s = 1 the normal brushed grain acquires crack-like dark/bright lines.

4.3 Sensitivity: gain and blur kernel

Setting	Edge overshoot (halo)	CNR	Noise amplification	False alarms / img	Recall
Unsharp/high-boost k = 0.5, s = 1	5.3 %	4.19	x1.43	0.25	63%
Unsharp/high-boost k = 0.5, s = 2	11.7 %	4.37	x1.49	0.35	72%
Unsharp/high-boost k = 0.5, s = 4	17.8 %	4.65	x1.50	0.30	72%
Unsharp/high-boost k = 1, s = 1	10.0 %	3.85	x1.86	0.10	59%
Unsharp/high-boost k = 1, s = 2	22.6 %	4.13	x1.97	0.20	65%
Unsharp/high-boost k = 1, s = 4	34.6 %	4.55	x1.99	0.40	72%
Unsharp/high-boost k = 3, s = 1	27.8 %	3.00	x3.58	0.05	54%
Unsharp/high-boost k = 3, s = 2	66.4 %	3.44	x3.89	0.10	57%
Unsharp/high-boost k = 3, s = 4	102.5 %	3.96	x3.96	0.40	70%
Unsharp/high-boost k = 5, s = 1	46.7 %	2.52	x5.30	0.00	50%
Unsharp/high-boost k = 5, s = 2	110.6 %	2.98	x5.80	0.00	57%
Unsharp/high-boost k = 5, s = 4	163.0 %	3.40	x5.91	0.40	70%
Laplacian c=0.5	10.8 %	3.02	x3.16	0.05	54%
Laplacian c=1.0	20.9 %	2.33	x5.36	0.00	43%
Laplacian c=2.0	42.4 %	1.71	x9.67	0.00	22%

Table 4.2 - Gain (k) and blur-sigma sensitivity on 20 images. Overshoot > 50 % is a clear halo.

Halo: with s = 4 px the mask is wide, so overshoot at the strong edge grows from 18 % (k = 0.5) to 163 % (k = 5): a bright/dark rim 4 px wide appears around every strong boundary. **False crack from grain:** counting local deviations of more than 12 grey levels on defect-free sheets (a fixed-threshold viewer rule), the original has 29 per image, but unsharp (k=1) gives 596, high-boost k = 3 gives 2358, Laplacian 3798 and k = 5 with s = 1 gives 2812 - normal grain streaks reach crack-like contrast (Fig. 4.2, bottom). The adaptive-threshold detector does not show this because it rescales its threshold to the amplified noise, which is exactly why a fixed visual criterion must not be used on a sharpened picture.

4.4 Analysis and recommendation

The Laplacian gives the narrowest scratch (FWHM 2 px, max gradient 72) but amplifies noise x5.4, and its PSNR falls from 38.4 to 22.9 dB with CNR falling to 2.3 and recall to 44 %. The high-boost is the deepest (53 grey levels) but has x3.8 noise amplification and 49 % overshoot. **Unsharp masking** with small gain is the best compromise (x1.9 noise, 17 % overshoot at k = 1; at **k = 0.5, s = 2: overshoot 11.7 %, noise x1.49, CNR 4.37 and recall 72 %, equal to the unsharpened image**).

> **Recommendation: unsharp masking with k = 0.5-1.0 and sigma = 1.5-2 px, used only on the operator display copy after denoising.** Sharpening is not applied before detection (Task 10), because every setting reduced or at best equalled the detection recall of the original (72 %). **A sharpened image must not replace the original evidence image** because it changes the shape and apparent width of defects (halo adds a false rim; FWHM changes), alters area and perimeter measurements, and can create or hide features; the original is archived untouched for 30 days and shown alongside.
>
> **Outcome vs prediction:** Confirmed in full: ranking Laplacian > high-boost > unsharp for both edge sharpness and noise amplification; halos and false cracks appear at high gain and small blur.

Task 5 - Frequency-domain enhancement: low-pass and high-pass filters [5 marks]

> **Prediction (formed from theory before running the code):** Brushed texture is a horizontal streak pattern whose energy lies along the vertical-frequency axis; slowly varying illumination lies at very low radius; thin scratches and pit boundaries occupy mid/high radius. An ideal (brick-wall) filter should give the strongest ringing, Butterworth less, Gaussian none. A low-pass filter alone should reduce noise but erase fine scratches; a band-pass (high-pass to remove illumination, then low-pass to remove noise) should give the best visibility.

5.1 Procedure

Each image (512 x 512) is transformed with np.fft.fft2, the zero frequency shifted to the centre, multiplied by a real filter mask H(u,v) built from the distance D from the centre, and reconstructed with ifft2. Masks: **ideal** H = 1 for D <= D0; **Butterworth** H = 1/(1+(D/D0)^2n) with n = 2 (and 4 for one test); **Gaussian** H = exp(-D^2/2 D0^2). High-pass = 1 - low-pass (the mean grey level is added back for display). Cut-offs are in cycles per image: **low-pass D0 = 60** (period 8.5 px = 1.3 mm: keeps everything coarser than the 2 px scratch cross-section) and **high-pass D0 = 15** (period 34 px = 5.3 mm: removes illumination variation across the field but keeps scratches and pits). A strong low-pass D0 = 15 and a Gaussian band-pass (HP D0 = 15 x LP D0 = 90) were also tested. **16 images** (26 pit/pinhole/scratch/crack instances) were used.

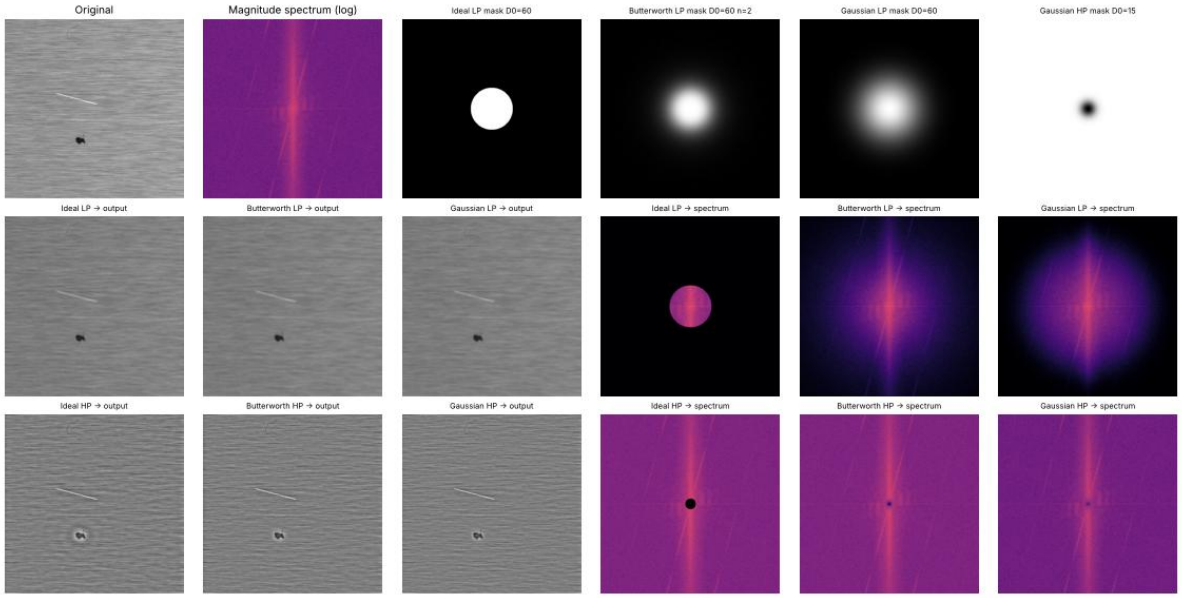


Figure 5.1 - Magnitude spectrum (log), the three low-pass masks and high-pass mask (top); low-pass outputs and their spectra (middle); high-pass outputs and their spectra (bottom). The streak texture appears as a vertical band through the centre of the spectrum.

5.2 Which structures are retained or attenuated

- **Low-pass (D0 = 60):** retains illumination gradient and coarse stains and the broad scale marks; attenuates sensor noise, grain and the high-frequency edge of 2 px scratches (their spectrum extends beyond D = 60), and the periodic fine streaks of the brushed finish.
- **High-pass (D0 = 15):** removes the gradient/highlight and vignetting (healthy-region standard deviation drops from 21.6 to 4.2-4.6 grey levels), and retains scratches, pit boundaries and texture; noise is retained (not amplified), but the relative weight of noise and texture increases.
- **Ideal vs Butterworth vs Gaussian:** the three masks have the same D0 but different transitions: ideal is a hard circle, Butterworth falls over a band around D0, Gaussian falls smoothly. The corresponding spatial kernels are a sinc (ideal, many side-lobes), a damped ringing kernel (Butterworth) and a Gaussian with no side-lobes.

5.3 Comparison

Filter	CNR (all)	CNR (low-contrast)	Recall	False detections / img	Healthy std (grey lv)	Time (ms)	Ringing (over/under)
Original	4.20	1.75	77%	0.8	21.6	0.0	-
Ideal LP D0=60	3.49	1.65	62%	18.1	20.9	8.9	9.1% / 5.3%
Butterworth LP D0=60	4.25	1.78	69%	12.1	20.9	11.9	3.4% / 2.8%
Gaussian LP D0=60	4.55	1.84	65%	9.2	20.8	11.3	0.0% / 0.0%
Ideal HP D0=15	4.77	2.14	73%	14.1	4.6	9.1	-
Butterworth HP D0=15	4.80	2.19	77%	11.0	4.3	10.2	-
Butterworth HP D0=15 n=4	4.87	2.22	73%	12.8	4.4	9.6	-
Gaussian HP D0=15	4.67	2.14	77%	9.7	4.2	10.4	-
Gaussian LP D0=15 (strong)	1.03	0.50	38%	9.8	20.2	12.9	-
Gaussian band-pass HP15 x LP90	5.81	2.73	73%	11.8	2.9	10.1	-

Table 5.1 - Defect visibility and false detections on 16 images. Ringing = overshoot / undershoot at a 60-grey-level step for low-pass D0 = 60 (D0 = 25 values in Fig. 5.2). False detections use the unchanged 4-sigma rule, which is not re-tuned for each filter, so the column is a relative indicator.

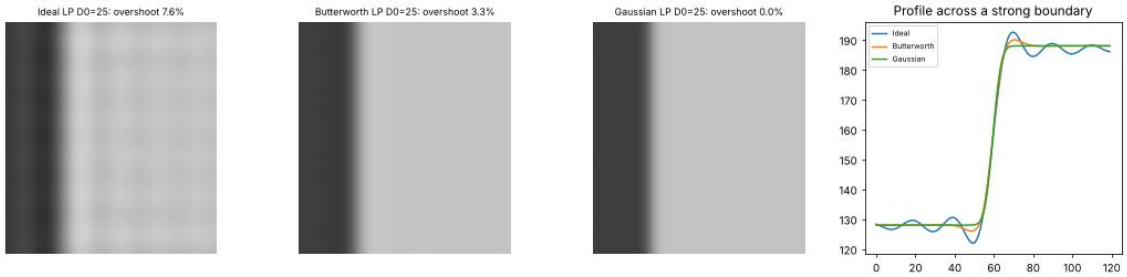


Figure 5.2 - Ringing near a strong boundary (low-pass D0 = 25): the ideal filter produces visible overshoot (7.6 %) and undershoot (10.1 %); Butterworth n = 2 3.3 %; Gaussian none.

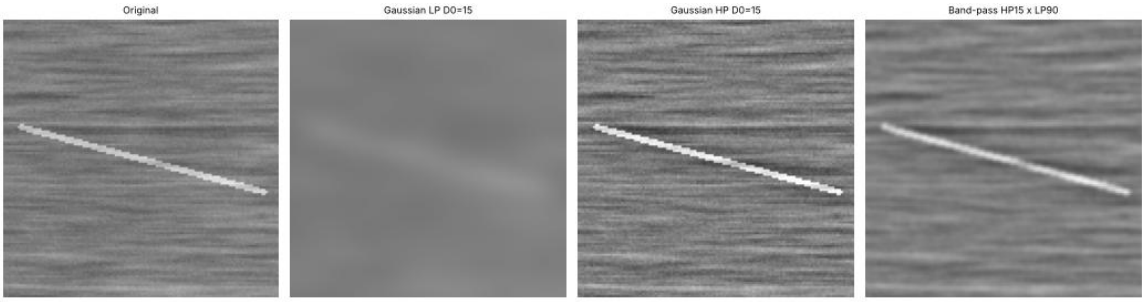


Figure 5.3 - A scratch crop: original, Gaussian low-pass D0 = 15 (scratch erased), high-pass and band-pass (scratch and pits emphasised, illumination removed).

5.4 Analysis

- **Ringing:** the ideal filter shows 9.1 % overshoot / 5.3 % undershoot at D0 = 60 and 7.6 % / 10.1 % at D0 = 25; Butterworth n = 2 about 3 %; Gaussian 0 %. Ringing from the ideal low-pass produces parallel ghost lines that could be mistaken for hairline cracks beside a strong boundary.
- **Edge preservation and noise:** Gaussian LP D0 = 60 has the highest CNR of the low-pass set (4.55 vs 4.25 Butterworth and 3.49 ideal; the ideal low-pass is below the original 4.20 because ringing adds structured error), and the lowest false-detection rate (9.2 vs 12.1 and 18.1).
- **Contrast:** high-pass filters raise CNR by 11-16 % over the original (4.67-4.87 vs 4.20) and low-contrast CNR from 1.75 to 2.14-2.22, because removing the illumination gradient reduces the ring-std denominator. The **Gaussian band-pass gives the highest CNR (5.81, +38 %) and low-contrast CNR (2.73, +56 %)**.

- **Why a low-pass filter alone cannot detect fine scratches:** a 2 px wide line has most of its energy at spatial frequencies above 0.1 cycles/px. A Gaussian low-pass with D0 = 15 (0.03 cycles/px) reduces CNR from 4.20 to 1.03 and recall from 77 % to 38 %; even D0 = 60 reduces recall to 65 %. Low-pass filtering suppresses exactly the band in which the defects are; its legitimate use is to estimate the illumination/background and to denoise, followed by a high-pass or subtraction.
- **Time:** 9-13 ms per 512 x 512 image with NumPy FFT (single core), independent of the filter type.

5.5 Recommendation

Gaussian frequency filters in sequence: Gaussian high-pass (D0 = 15 cycles/image, about 5 mm) to remove illumination, then Gaussian low-pass (D0 = 90) to limit noise (a Gaussian band-pass). It gives the best measured visibility, no ringing and one fixed 10 ms cost. Butterworth (n = 2) is an acceptable alternative where a sharper cut-off is needed; the ideal filter is not used because of ringing next to sheet edges and strong marks. For the final pipeline the same effect is obtained by a Difference-of-Gaussians in the spatial domain (about 12 ms), which avoids wrap-around errors of the FFT at the image border.

Limitation: the false-detection column mixes enhancement with a threshold not tuned to each filter; the FFT assumes periodic image borders (a 20 px border margin is ignored in real use); the cut-off values are matched to 0.156 mm/px and must be rescaled if the optics change.

Outcome vs prediction: Confirmed: ringing ranking ideal > Butterworth > Gaussian; low-pass alone destroys scratches; band-pass is best. The ideal high-pass had slightly lower recall (73 %) than Butterworth and Gaussian (77 %), consistent with the ringing penalty.

Task 6 - Point and line detection with spatial masks [5 marks]

Prediction (formed from theory before running the code): A Laplacian point mask will detect pits and pinholes and, with weaker response, scratch segments; it will also respond to dust, so false positives from dust are expected. A single directional mask responds fully only to lines within about 22 degrees of its direction and nearly zero to the orthogonal direction; taking the maximum of four orientations should recover diagonal scratches, but will also respond to the horizontal brushed texture.

6.1 Mask definitions (cv2.filter2D)

Mask	Coefficients (3 x 3)	Meaning / response sign
Point (Laplacian-type)	[-1 -1 -1; -1 8 -1; -1 -1 -1]	Coefficients sum to 0, so a flat area gives 0. Response R = 8 f_c - sum(neighbours): strongly positive for a bright point, strongly negative for a dark point. Rule: |R| > T.
Horizontal line	[-1 -1 -1; 2 2 2; -1 -1 -1]	Sum 0; responds positively to a bright horizontal line 1 px wide (2+2+2 centre row against -1 above/below). A dark line gives a negative response -> use |R|.
Vertical line	[-1 2 -1; -1 2 -1; -1 2 -1]	Transpose of the horizontal mask.
Diagonal +45 deg and -45 deg	[-1 -1 2; -1 2 -1; 2 -1 -1] and [2 -1 -1; -1 2 -1; -1 -1 2]	Weights of 2 on the diagonal; same sign convention.
Multi-orientation (C)	max over the four line masks of |R|	Isotropic-ish response to lines of any angle.

Threshold rule: after a 0.8 px Gaussian pre-smoothing, a pixel is declared when |R| exceeds median(|R|) + k * sigma_R, sigma_R being the robust (MAD) spread of the response in that image. **k is chosen per strategy on four defect-free images that are not used for testing**: the smallest k that gives at most 0.5 false components per image, so normal rolling marks and grain are below threshold by construction (k = 7 point, 8 horizontal, 30 vertical, 20 diagonal, 10 multi-orientation). Components smaller than 3 px are discarded.

6.2 Synthetic patterns

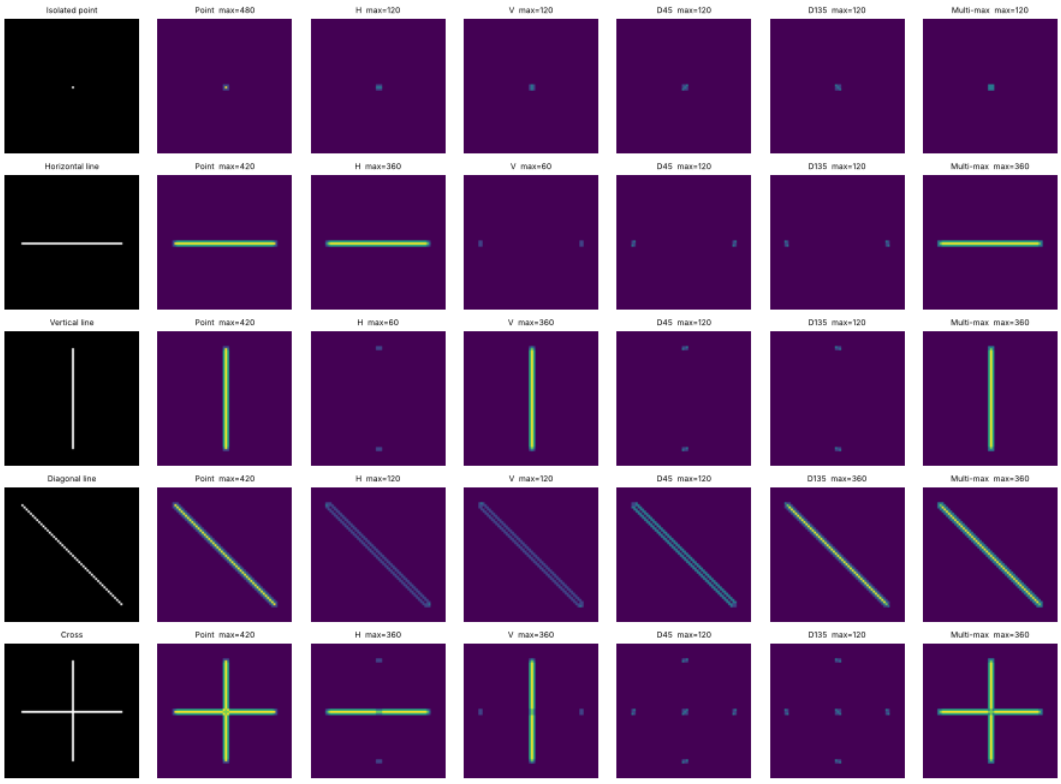


Figure 6.1 - Five synthetic patterns (bright, 1 px wide, contrast 60) and the absolute response of each mask (columns: pattern, point, H, V, +45, -45, multi-orientation).

Pattern	Point	H	V	+45	-45	Multi-max
Isolated point	480	120	120	120	120	120
Horizontal line	420	360	60	120	120	360
Vertical line	420	60	360	120	120	360
Diagonal line	420	120	120	120	360	360
Cross	420	360	360	120	120	360

Table 6.1 - Peak |response| (ideal values: point 480 for the point mask = 8 x 60; line 360 = 6 x 60). The orthogonal mask gives 60, the others 120.

6.3 Real/representative metal images

22 images (12 with defects, 4 defect-free, 6 sheet-boundary images) with manual reference labels: 8 point-like defects (pits/pinholes) and 19 line-like defects (scratches/cracks). True detections are defects touched by a detected component; false positives are components touching no defect.

Strategy	Threshold k	Recall points (n=8)	Recall lines (n=19)	Precision	False positives / img	Time (ms)
A Point (Laplacian)	7	63%	68%	71%	2.09	12.9
B Horizontal mask only	8	38%	47%	80%	0.50	14.2
B Vertical mask only	30	38%	21%	36%	0.82	15.0
B Diagonal +45 mask only	20	38%	47%	97%	0.14	14.8
C Multi-orientation (max of 4)	10	38%	47%	65%	1.14	15.4

Table 6.2 - Detection performance (22 images; threshold calibrated on separate defect-free images).

Strategy	False-positive sources (count over 22 images)	Scratch recall by orientation (n)
A Point (Laplacian)	dust 9, grain/texture 6, sheet boundary 31	horizontal 67% (3), diagonal 67% (6), vertical 67% (3)
B Horizontal mask only	dust 6, grain/texture 5	horizontal 67% (3), diagonal 33% (6), vertical 0% (3)
B Vertical mask only	dust 9, sheet boundary 9	horizontal 0% (3), diagonal 0% (6), vertical 67% (3)
B Diagonal +45 mask only	dust 3	horizontal 33% (3), diagonal 50% (6), vertical 33% (3)
C Multi-orientation (max of 4)	dust 6, grain/texture 5, sheet boundary 14	horizontal 67% (3), diagonal 33% (6), vertical 33% (3)

Table 6.3 - False-positive categories (dust, rolling mark, sheet boundary, grain) and orientation dependence.

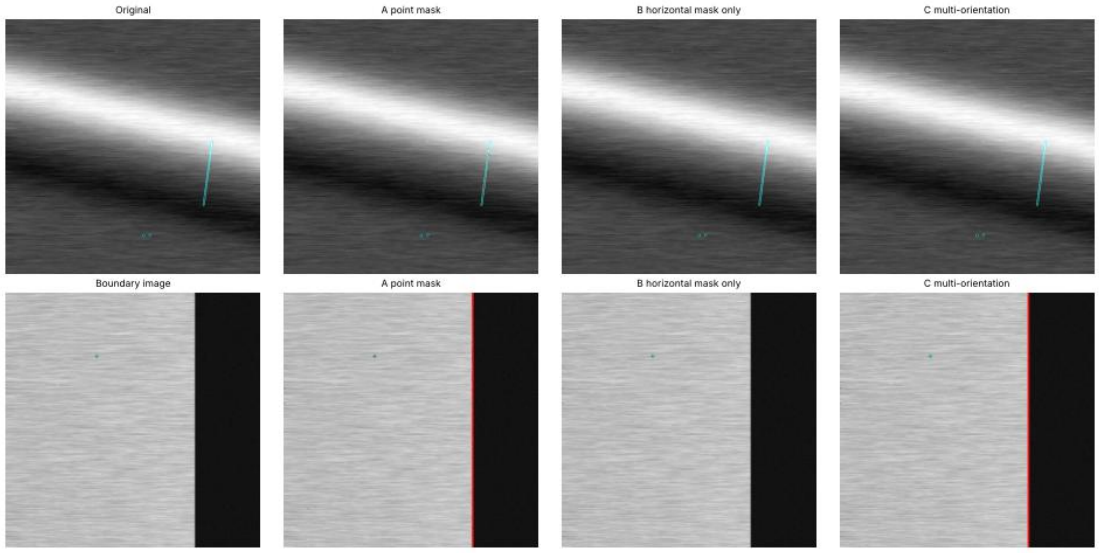


Figure 6.2 - Detections (red) over labelled defects (cyan) for the point mask, the horizontal-only mask and the multi-orientation response. The second row shows the response at the sheet boundary.

6.4 Orientation dependence

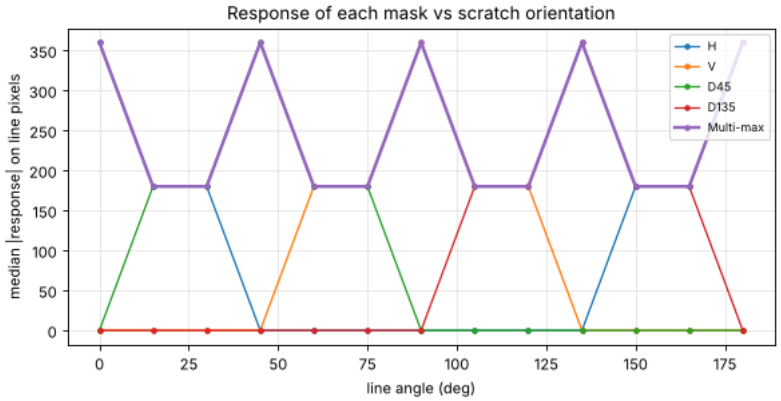


Figure 6.3 - Median |response| on a 1 px line of contrast 60 rotated from 0 to 180 degrees.

A single directional mask is matched to one angle. The horizontal mask gives its full response (360) at 0 and 180 degrees, **half (180) at 15, 30, 150 and 165 degrees and essentially zero for 45-135 degrees** (Fig. 6.3). A diagonal scratch at 45 degrees crosses the horizontal mask as a one-pixel-wide diagonal: the positive 2-weights meet only one pixel per row against -1 weights over the rest, so the sum over the mask is about zero. Using one mask therefore **misses a diagonal scratch** - in the data the horizontal-only mask detected 67 % of horizontal scratches, 33 % of diagonal and 0 % of vertical ones. The maximum over the four masks has a worst case (15 and 30 degrees) of 50 % of the ideal response, so even four masks leave a gap between orientations; a finer set (eight orientations or oriented Gabor/steerable filters) would fix this at higher cost.

6.5 Analysis and recommendation

- Point mask: highest recall for both pits (62 %) and, helped by their short bright/dark runs, line segments (68 %) at 0.71 precision; false positives are mostly dust and the sheet boundary (31 of 42 components in the boundary images are at the edge).
- Multi-orientation: at the calibrated k = 10 it recovers only 47 % of lines at 65 % precision; the horizontal mask also responds to the horizontal brushed texture, which is why its required k is high. At a low k (5) recall rises to 95 % but there are 150 false components per image, almost all from texture and rolling marks (Table 6.4).

Threshold k (multi-orientation)	Recall points	Recall lines	Precision	False components / img
5.0	63%	95%	3%	149.6
7.5	63%	63%	27%	11.1
10.0	38%	47%	65%	1.1
15.0	25%	21%	19%	5.3
20.0	13%	16%	33%	0.5

Table 6.4 - Threshold sweep for the multi-orientation response: the usable operating range is narrow because rolling marks and grain are in the same response range as faint scratches.

Recommended mask strategy: run the **point mask (A) and the multi-orientation response (C) in parallel as a candidate generator**, each at the threshold calibrated on defect-free images (and re-calibrated whenever the camera, lens or lighting changes). Candidates are then judged in the later stages by length, orientation and contrast: rolling marks are long, nearly horizontal and low contrast, whereas defects are short or at other angles, so a candidate is rejected as a rolling mark only if its length exceeds the rule in Task 10. The threshold is **set from the response distribution of known-good sheets** (just above the maximum response of normal grain and rolling marks), never from defective samples, and stored per sheet type (stainless vs mild steel).

Limitation: 3 x 3 masks are matched to 1 px lines; 2 px, low-contrast scratches (CNR below 2) fall below the noise floor of any 3 x 3 mask at an acceptable false-positive rate, so recall above 70 % was not reached. The reference set has only 8 point-like and 19 line-like defects.

> **Outcome vs prediction:** Confirmed: single directional masks miss other orientations; dust and boundaries cause false positives; point mask responds to line fragments. Not confirmed: I expected the multi-orientation response to beat the point mask in recall - it did not at equal false-positive budget, because the horizontal mask is dominated by the horizontal texture.

Task 7 - Edge detection: Sobel, Prewitt and Canny [4 marks]

> **Prediction (formed from theory before running the code):** Sobel and Prewitt are nearly identical (Sobel weights the centre row more strongly, giving slightly better noise behaviour); both produce thick, broken edges and must be thresholded globally. Canny gives thinner, better localised and more continuous edges, and hysteresis should keep weak crack segments that touch strong ones while rejecting grain; its two thresholds control broken edges (too high) and false edges (too low).

7.1 Implementation and parameters

All three operate on the same 20 evaluation images (a separate set of 8 images was used to tune thresholds), after the same Gaussian smoothing (sigma 1.2). **Sobel:** cv2.Sobel 3 x 3, magnitude |Gx| + |Gy|. **Prewitt:** kernels [[-1,0,1],[-1,0,1],[-1,0,1]] and the transpose via cv2.filter2D, same magnitude. Both are thresholded at median + k sigma (robust spread of the magnitude) and thinned with skeletonisation so that their edges can be compared fairly with Canny. **Canny:** cv2.Canny, aperture 3, low/high thresholds at median + a sigma and median + b sigma. Parameters (k, a, b) were chosen by grid search for the best F1 on the tuning images: **Sobel k = 6, Prewitt k = 6, Canny (a, b) = (2, 8)**. Reference edges: boundaries of the manually labelled defect masks (stain edges are ignored because they are blurred and have no true edge); tolerance 2 px.

7.2 Results

Operator	Precision	Recall	F1	Recall on weak (low-contrast) edges	Fragments per crack	Edge pixels on defect-free sheets	Time (ms)
A Sobel	0.83	0.61	0.70	0.40	1.75	0.006 %	11.4
B Prewitt	0.83	0.59	0.69	0.39	0.75	0.006 %	12.6
C Canny	0.92	0.69	0.79	0.48	0.00	0.003 %	11.2

Table 7.1 - Edge-quality metrics on 20 images (2 px tolerance).

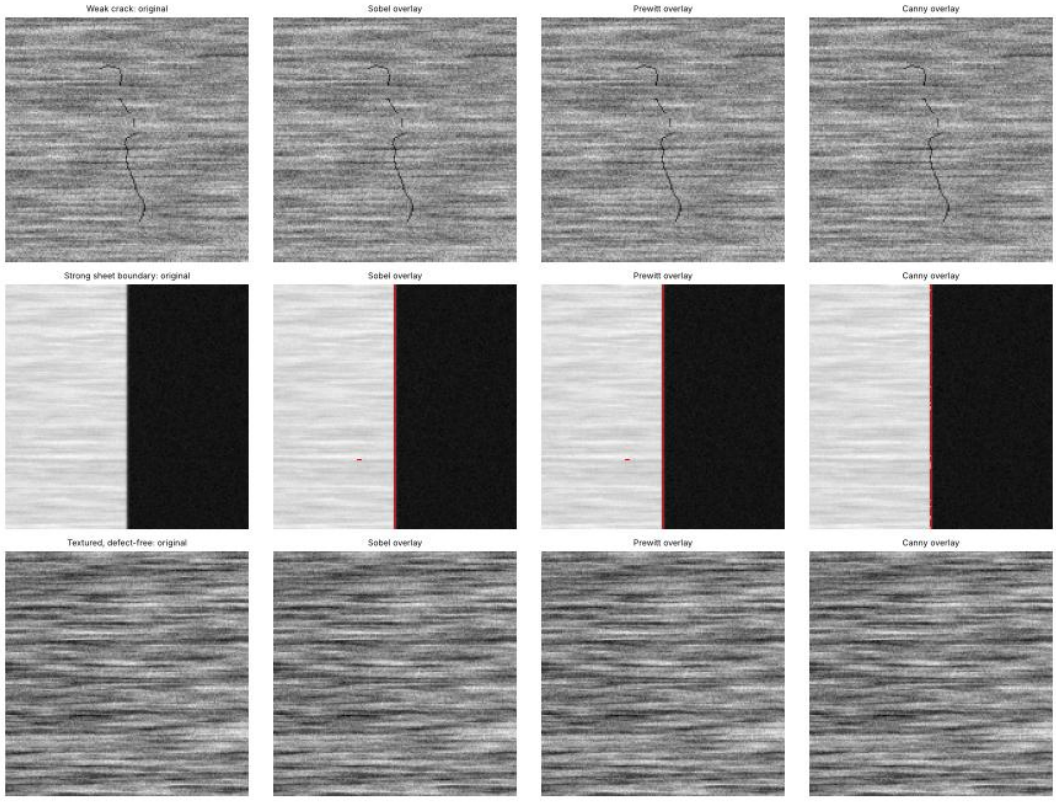


Figure 7.1 - Edge overlays (red) on original crops: a weak crack (top), a strong sheet boundary (middle) and a textured, defect-free region (bottom).

7.3 Effect of the Canny thresholds

Low / high (sigma units)	Precision	Recall	F1	Recall on weak edges	Fragments per crack	False edge density (%)
1 / 3	0.02	0.91	0.04	0.82	9.25	4.847
2 / 6	0.47	0.78	0.58	0.59	1.25	0.081
3 / 8	0.93	0.64	0.76	0.40	0.00	0.003
4 / 10	0.96	0.47	0.63	0.28	0.00	0.002
5 / 14	1.00	0.27	0.42	0.11	0.00	0.000
8 / 20	1.00	0.24	0.39	0.06	0.00	0.000

Table 7.2 - Canny threshold sensitivity.

- **Thresholds too low (1 / 3):** recall 0.91 and weak-edge recall 0.82, but precision 0.02 and 4.8 % of the pixels of a defect-free sheet become edges - the brushed grain is turned into edge noise; a crack is cut into 9 fragments because edge pixels wander.
- **Thresholds too high (5 / 14 and above):** precision 1.0 but recall 0.27 and weak-edge recall 0.11 - the crack boundary is broken or lost because the gradient along the crack is intermittently below the high threshold.
- **Balanced (2-3 / 6-8):** F1 0.76-0.79; hysteresis keeps weak segments connected to strong ones: 0 fragments per detected crack compared with 1.75 (Sobel) and 0.75 (Prewitt).

7.4 Analysis and recommendation

Sobel and Prewitt behave alike (F1 0.70 and 0.69); both give thick ridge responses that need thinning and give separate fragments along weak cracks. **Canny is best for every measured aspect:** F1 0.79, precision 0.92, recall 0.69, weak-edge recall 0.48, continuity, the lowest density on grain (0.003 % versus 0.006 %), at the same 12 ms cost. **Recommended operator for the next segmentation stage: Canny (sigma 1.2, thresholds at median + 2 sigma and median + 8 sigma of the Sobel gradient magnitude),** used to provide boundaries for region linking and validation, not as the only detector.

Limitation: even the best operator finds only 48 % of weak edges because a 6-12 level crack boundary has a gradient comparable to the grain gradient; Canny responds to the strong **sheet boundary** (middle row of Fig. 7.1) and to the edges of scale marks, so a region-of-interest mask must exclude the sheet boundary; the thresholds are relative to the image noise level and must be checked whenever the camera gain changes.

Outcome vs prediction: Confirmed: Sobel and Prewitt are near-equivalent, Canny is best and its thresholds trade broken edges against false responses.

Task 8 - Thresholding: fixed, Otsu and adaptive [5 marks]

> **Prediction (formed from theory before running the code):** A fixed global threshold will work only on normal illumination and flag huge areas on dim or uneven images. Otsu will also fail because defects occupy less than 1 % of the pixels so the histogram is not bimodal; it can work only when a large defect makes it bimodal. Adaptive thresholding will cope with illumination but will lose low-contrast defects and may create false regions in textured healthy metal if the constant C is too small.

8.1 Procedure and polarity

Polarity convention: defects are dark foreground (mask = 1 where the pixel is darker than its surroundings); bright defects (bright scratches, bright stains) are handled in Task 10 by running the same method on the inverted image and are not scored in this task. After a Gaussian smoothing of 1 px, three methods were run on **40 images** (all conditions, including 8 normal images) and every mask is cleaned the same way (components smaller than 6 px removed). The reference is the union of the dark defects. A: fixed T = 112 (cv2.threshold), chosen as about 130 - 4.5 sigma on a normally lit image. B: THRESH_OTSU on the image, and B' = Otsu on the flat-fielded image (background subtracted, sigma 25). C: cv2.adaptiveThreshold, Gaussian-weighted mean, block 31, C = 8.

8.2 Segmentation quality

Method	Precision	Recall	IoU	Dice	Recall, low-contrast defects	Image area flagged on defect-free sheets	Time (ms)
A Fixed global (T=112)	0.003	0.51	0.003	0.006	0.46	53.0 %	2.2
B Otsu (raw image)	0.005	0.92	0.005	0.009	0.96	51.6 %	2.5
B' Otsu on flat-fielded image	0.004	0.97	0.004	0.008	0.97	53.3 %	14.2
C Adaptive Gaussian (block 31, C=8)	0.607	0.30	0.249	0.399	0.23	0.0 %	3.3

Table 8.1 - Pixel-level comparison with the reference masks, 40 images (pixels summed over the set).

Method	IoU normal	IoU dim	IoU uneven	IoU reflective
A Fixed global (T=112)	0.499	0.002	0.002	0.004
B Otsu (raw image)	0.006	0.005	0.003	0.005
B' Otsu on flat-fielded image	0.005	0.004	0.003	0.005
C Adaptive Gaussian (block 31, C=8)	0.355	0.415	0.250	0.110

Table 8.2 - IoU by capture condition. The fixed threshold works on normally lit sheets (IoU 0.50) and fails on every other condition.

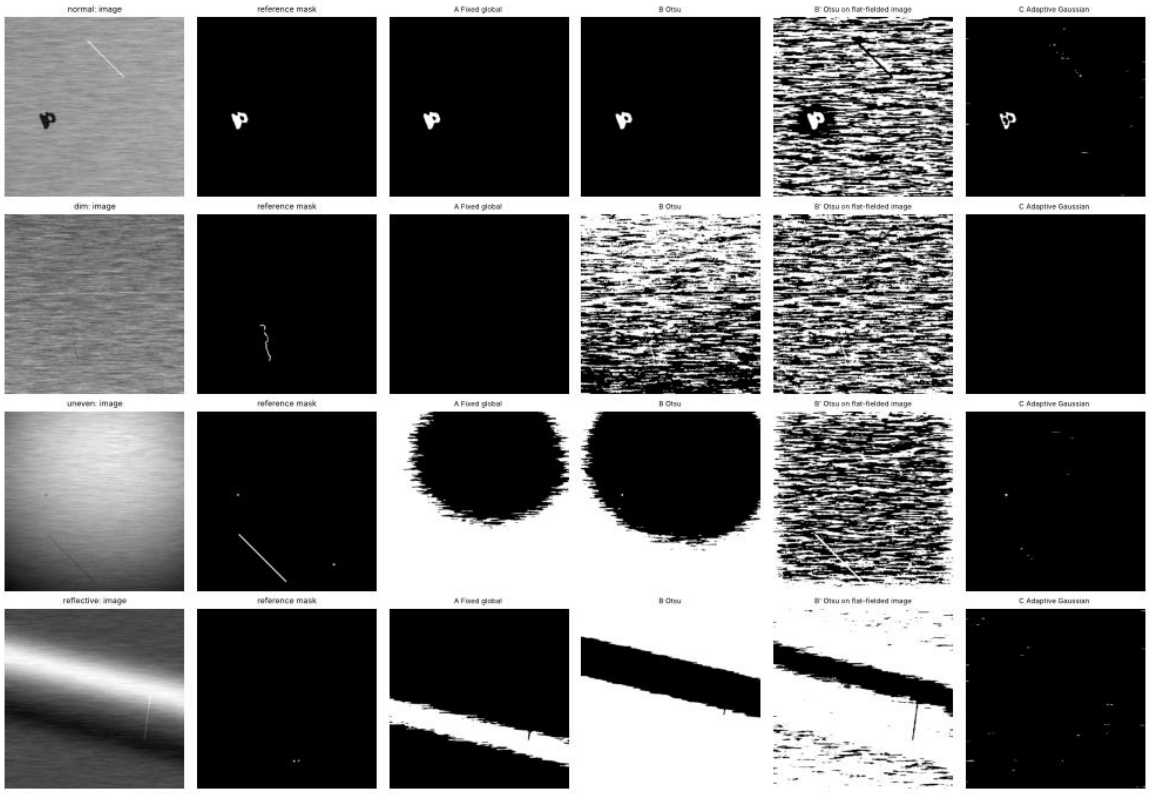


Figure 8.1 - Binary masks for a normal, dim, unevenly lit and reflective sheet (rows): image, reference, fixed, Otsu, Otsu on flat-fielded image, adaptive.

8.3 Under- and over-segmentation: parameter variation

Fixed threshold T	Precision	Recall	IoU	Measured/true defect area	Defect-free area flagged
90	0.003	0.39	0.003	x135.2	45.0 %
100	0.003	0.43	0.003	x151.6	48.4 %
110	0.003	0.50	0.003	x170.3	52.2 %
112	0.003	0.51	0.003	x174.0	53.0 %
120	0.003	0.59	0.003	x190.3	56.4 %
130	0.003	0.79	0.003	x274.6	74.5 %
140	0.002	0.86	0.002	x374.6	95.2 %

Table 8.3 - Fixed-threshold sweep. A low T under-segments (recall 0.39 at T = 90); a high T over-segments and the measured area becomes x135-375 the true area - a size-based reject decision would be wrong on every sheet.

Block size	Constant C	Precision	Recall	IoU	Measured/true area	Defect-free area flagged
11	4	0.146	0.24	0.099	x1.63	0.23 %
11	8	0.991	0.07	0.068	x0.07	0.00 %
11	12	1.000	0.01	0.006	x0.01	0.00 %
31	4	0.026	0.47	0.025	x18.16	3.36 %
31	8	0.607	0.30	0.249	x0.49	0.03 %
31	12	0.993	0.22	0.215	x0.22	0.00 %
61	4	0.021	0.61	0.021	x28.52	5.30 %
61	8	0.296	0.37	0.198	x1.26	0.15 %
61	12	0.960	0.33	0.326	x0.34	0.00 %

Table 8.4 - Adaptive threshold sweep. C = 4 over-segments texture (area x18-29, 3-5 % of a normal sheet flagged); C = 12 gives precision above 0.96 but under-segments (area x0.2-0.34). Best IoU 0.33 at block 61, C = 12.

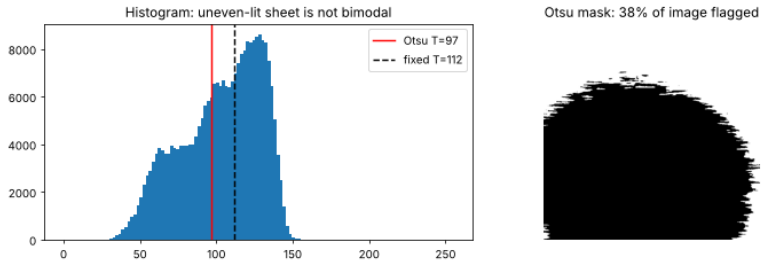


Figure 8.2 - Why Otsu fails: a histogram that is a single broad peak (illumination spread, no second mode); Otsu splits it in the middle and flags 38 % of the sheet.

8.4 Why Otsu fails and why adaptive thresholding can produce false regions

Otsu chooses the threshold that maximises the between-class variance. When defects are less than 1 % of the pixels the histogram is a single broad peak (healthy texture plus illumination spread), the "two classes" are just the two halves of that peak, and Otsu returns a threshold at the middle - **it flagged 52 % of a defect-free sheet on average** (51.6 % raw, 53.3 % on the flat-fielded image) and for an unevenly lit sheet 38 % (T = 97), for a reflective sheet 79 % (T = 173). In images where a large dark scale mark makes the histogram bimodal (for example a normal-illumination image with T = 105 and 0.5 % flagged against 0.47 % true) Otsu works - the method is correct only when the histogram has two modes, which cannot be guaranteed on line. **Adaptive thresholding** compares each pixel with its local mean minus C. On healthy brushed metal the local texture has a standard deviation of about 4 grey levels; if C is only 1-2 sigma, many texture pixels fall below the local mean - C and create small false regions (3-5 % of a defect-free image at C = 4). A block that is small relative to the defect (block 11) also adapts to the defect itself and erases it, which is the cause of its poor recall.

8.5 Recommendation and mask check before any reject decision

Recommended: a controlled combination - illumination flattening, then a local (adaptive) threshold with hysteresis (a strong threshold to seed regions and a weaker threshold to grow them, implemented in Task 10), with polarity handled by processing the dark and bright sides separately. Of the three simple methods only adaptive thresholding is usable (IoU 0.25-0.33 compared with 0.003-0.005); a fixed threshold is kept only as a sanity check for grossly wrong exposure (for example if more than 20 % of the image is below T, the system reports "illumination fault" rather than "defects").

Checking the binary mask before a reject decision: (1) reject the mask automatically as unreliable if the flagged area exceeds a plausibility limit (for example 2 % of the sheet area) or the number of regions exceeds a set count; (2) remove regions under 6 px (about 0.15 mm2) and regions that coincide with the saturated-highlight mask; (3) show the mask outline over the **original** image for operator confirmation if the region is close to the acceptance limit; (4) never reject on a single mask pixel group without the region measurements of Task 9.

Limitation: IoU is low even for the best method (0.33) mainly because 2 px wide scratches have a large perimeter/area ratio so a 1 px boundary error halves the IoU, and because low-contrast defects (recall 0.23 at the default setting) are not separable at an acceptable false-alarm rate.

Outcome vs prediction: Confirmed: fixed and Otsu thresholds fail outside normal illumination (IoU about 0.003-0.005 against 0.25-0.33); Otsu fails by splitting a unimodal histogram; adaptive thresholds trade false regions (small C) against lost defects (large C).

Task 9 - Region-based segmentation, edge linking and boundary detection [6 marks]

Prediction (formed from theory before running the code): Region growing from strong seeds will recover cracks as one region if the similarity tolerance is chosen well, but can leak or stop at gaps; connected-component analysis of a thresholded mask will fragment a hairline crack (gaps) and will merge touching defects; edge linking will close small gaps but, if the linking distance is larger than the distance between defects, will merge separate defects.

9.1 Implementation

The three approaches were applied to the same **20 images** (including 4 images with two pits only 13 px apart, and 6 cracks that contain 1-2 px gaps). They work on a detection score map S that combines the flattened blob z-score (flat-fielded image divided by the noise sigma and 4.5) and the Task 6 multi-orientation line response (divided by its calibrated threshold), so S >= 1 is a candidate. Reference counts include pits, pinholes, scratches, cracks and scale marks (**40 instances**, 6 of them cracks); stains are excluded here because they are handled by the large-area channel of Task 10.

Approach	Parameters and rule
A Region growing	Seeds: pixels with S >= 1.1, processed strongest first (at most 2500). Neighbourhood: 8-connected (cv2.floodFill, FLOODFILL_FIXED_RANGE). Similarity: a neighbour joins if its score is at least max(0.55, 35 % of the seed score), i.e. the region extends to about 2.5 sigma or a third of the seed contrast. Regions under 6 px are dropped.
B Connected-region analysis	Binary mask S >= 1.0, components under 6 px removed, 8-connected labelling (cv2.connectedComponentsWithStats) and contours (cv2.findContours).
C Edge linking and boundary completion	Canny edges (Task 7), edge fragments dilated with a 7 x 7 elliptical structuring element, so gaps up to about 6 px (0.9 mm) are bridged, labelled as groups; the original edge pixels and filled interiors form the region.

9.2 Count and over/under-segmentation results

Approach	Reference defects	Detected regions	Matched to a defect	False regions	Over-segmentation (extra fragments)	Under-segmentation (merged regions)	Missed defects	Time (ms)
A Region growing	40	89	62	27	34	0	12	33.1
B Connected-region analysis	40	74	43	31	18	0	15	34.6
C Edge linking + boundary completion	40	38	27	11	2	1	14	11.3

Table 9.1 - Manually verified reference counts versus detected counts, 20 images.

Approach	Cracks recovered as one region	Cracks fragmented	Cracks missed	Fragments per detected crack	Close pit pairs resolved (n=4)	Rolling-mark false regions
A Region growing	33%	33%	33%	2.5	1 / 4	2
B Connected-region analysis	0%	17%	83%	2.0	1 / 4	4
C Edge linking + boundary completion	17%	0%	83%	1.0	0 / 4	0

Table 9.2 - Behaviour on the four required cases: broken crack, two nearby pits, scratch-like texture (rolling marks). Touching defects are shown in the synthetic test below.



Figure 9.1 - Region labelling around a broken crack (top) and two nearby pits (bottom): reference, A, B, C and raw Canny edges (random colour per region).

9.3 Edge linking: closing a gap and excessive linking

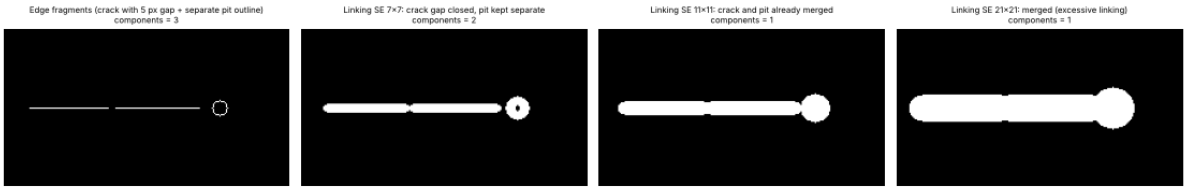


Figure 9.2 - Synthetic test: a crack with a 5 px gap and a separate pit outline 10 px beyond its end. Edge fragments give 3 components; linking with a 7 x 7 element closes the gap (2 components, defects still separate); 11 x 11 or larger merges the crack and the pit (1 component).

Linking element (px)	Cracks: extra fragments (real images)	Merged regions (real images)	Components in synthetic test (3 = no linking)
1 x 1	7	0	3
3 x 3	5	0	-
5 x 5	4	0	3
7 x 7	2	1	2
11 x 11	1	1	1
15 x 15	0	1	1
21 x 21	0	1	1

Table 9.3 - Linking-size sensitivity: extra fragments fall from 7 (no linking) to 0 at 15 x 15, while merged regions rise from 0 to 1 once the element exceeds the 6-10 px spacing.

9.4 Region properties in physical units

Calibration: **0.156 mm per pixel** (80 mm / 512 px) so 1 px2 = 0.0244 mm2. Contour perimeter from cv2.arcLength; shape measures are circularity 4 pi A / P^2 (1 = circle, capped at 1 for regions of a few pixels) and aspect ratio of the minimum-area rectangle.

Defect (example image)	Area (mm2)	Reference area (mm2)	Perimeter (mm)	Bounding box w x h (mm)	Circularity	Aspect ratio	Regions
pit (img 19)	0.29	0.49	1.5	0.6 x 0.6	1.00	1.0	1
pinhole (img 10)	0.34	0.15	1.8	0.6 x 0.6	1.00	1.0	1
scratch (img 7)	1.49	9.45	8.7	0.8 x 4.1	0.25	9.5	1
crack (img 11)	0.54	6.49	6.6	5.8 x 2.2	0.15	10.7	2
scale (img 0)	9.67	8.86	12.9	4.4 x 3.3	0.73	1.3	1

Table 9.4 - Measurements of one detected defect per type. The pit region (0.29 mm2, diameter about 0.6 mm) matches the 0.5 mm design size; the scratch measures 0.78 x 4.1 mm for the detected part of a 9.4 mm2 reference because only

the stronger segment was detected, which is why the measured area must always be accompanied by the detection confidence.

9.5 Analysis and recommendation

- **Region growing (A)** has the best recall (28 of 40 recovered, 12 misses) and can adapt the extent to the seed strength, but it produces 34 extra fragments (cracks split because the weak gap pixels fail the similarity rule) and costs 33 ms.
- **Connected-region analysis (B)** is simple and fast but a hairline crack yields about 2 fragments and 83 % of the cracks are missed at the calibrated threshold; touching defects always merge (a pure threshold has no way to separate them); 31 false regions.
- **Edge linking (C)** has the fewest false regions (0.55 per image, none from rolling marks) and the smallest over-segmentation (2 extra fragments), but it misses 14 defects (weak edges) and merged one pair of pits (0 of 4 close pairs resolved) - exactly the failure caused by the 7 x 7 linking element being larger than the 5 px gap between the two pits.

> **Recommended approach: connected-region analysis on the hysteresis mask (strong seeds, weak growth, which is the region-growing principle implemented with labelling) followed by boundary completion limited to a 5 x 5 element (gap <= 4 px, 0.6 mm), which is smaller than the 5 px gap between the closest defect pair.** The region is then measured in millimetres (Table 9.4) for the accept/reject rule.

Uncertain or merged regions: a region is sent for operator review (shown on the original image with the outline) if (i) the linked region contains two or more separately strong cores (possible merge of two defects), (ii) its area is within +/-30 % of the reject limit, (iii) it consists of fragments joined only by linking, or (iv) it overlaps the saturated highlight mask. Failure cases to show the operator: crack gap greater than the linking size (reported as two defects), close pits (merged), and weak cracks (missed).

> **Outcome vs prediction:** Confirmed: connected analysis fragments broken cracks; excessive linking merges separate defects; region growing reaches the best recall. Not confirmed: I expected region growing to recover cracks as single regions - only 33 % were, because the similarity criterion stops at the 1-2 px gaps.

Task 10 - Integrated application: enhancement and segmentation pipeline [8 marks]

Prediction (formed from theory before running the code): The basic pixel pipeline (A) will fail on dim, uneven and reflective sheets because a single global threshold cannot follow illumination. Pipeline B (band-pass + Canny + region validation) will have high precision but miss faint defects. The adaptive multi-stage pipeline (C) should have the best recall and IoU, at a higher time cost and with some extra uncertain regions; removing illumination flattening should hurt it most.

10.1 The three pipelines (executable Python/OpenCV code in the archive)

Pipeline	Stages
A Basic pixel pipeline	gray -> linear stretch (0.1/99.9 percentile) -> 3 x 3 median -> one global threshold pair, median +/- 4.5 x global robust sigma -> components >= 4 px.
B Enhancement + edge/region	Gaussian band-pass in the frequency domain (HP D0 = 15, LP D0 = 90) -> threshold |response| > 4.5 sigma -> remove regions < 8 px -> keep regions that contain a Canny edge pixel (sigma 1.2; 2 / 8) -> region measurements.
C Adaptive multi-stage	1 condition analysis (mean, saturation) -> 2 CLAHE for dim sheets -> 3 flatten (subtract Gaussian sigma 25) + denoise (sigma 1) + blob score -> 4 multi-orientation line response -> 5 hysteresis threshold (strong 1.15, weak 0.9 of calibrated unit) -> 6 boundary linking (dilate 5 x 5, label) -> 7 region validation and decision (area >= 6 px; rolling-mark rule; saturated-region rule; reject if score >= 1.6 and area >= 10 px else uncertain) -> 8 large-area (stain) channel on a 4x down-sampled image.

The parameters of C (strong 1.15, weak 0.9) were chosen by a 4 x 3 grid on the first 30 images by maximising recall minus 0.12 x false alarms per image (grid in the code archive), which gave recall 0.61, 1.6 false alarms per image on that subset. All three pipelines were then evaluated on the **same 60 labelled images (107 defect instances)**; the original image, enhanced image, binary mask and region annotations (with status reject/uncertain) are saved for each.

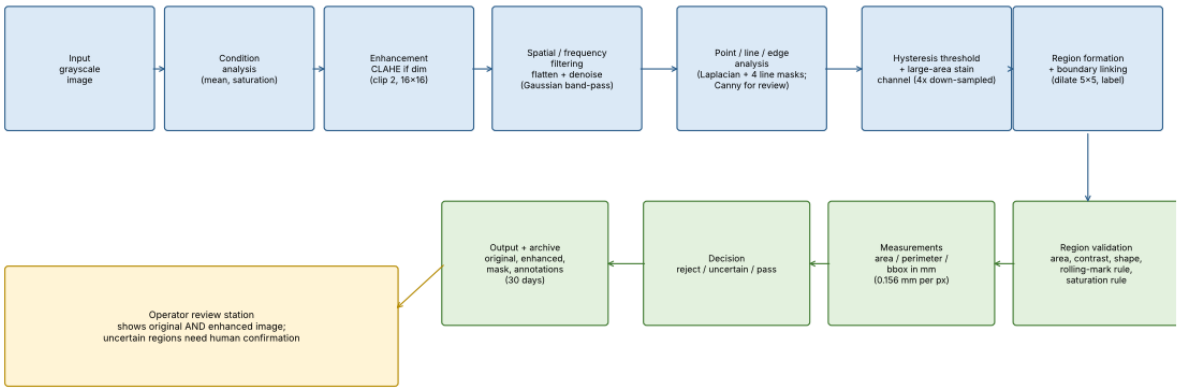


Figure 10.1 - End-to-end block diagram of the recommended pipeline C, including the review station and the 30-day archive.

10.2 Performance comparison

Pipeline	Defect recall	Region precision	IoU	Dice	False alarms / image	Missed defects (of 107)	Recall, low-contrast	Recall, reflective sheets	FA/img, reflective	Time (ms)
A Basic pixel	28%	12%	0.024	0.047	6.15	77	20%	9%	1.47	9.3
B Enhancement + edge/region	50%	89%	0.163	0.281	0.25	53	24%	35%	0.07	31.1
C Adaptive multi-stage	60%	46%	0.211	0.348	2.00	43	34%	32%	0.53	30.0

Table 10.1 - Common-dataset comparison. Defect recall: a defect counts as detected if a region touches it; false alarm: a region touching no reference defect. Time: 512 x 512 on one core.

Pipeline	normal: recall / FA per img	dim	uneven	reflective
A pipeline	30% / 0.07	87% / 23.07	0% / 0.00	9% / 1.47
B pipeline	52% / 0.07	65% / 0.40	56% / 0.47	35% / 0.07
C pipeline	78% / 0.53	83% / 0.67	59% / 6.27	32% / 0.53

Table 10.2 - Recall / false alarms per image by capture condition (15 images each).

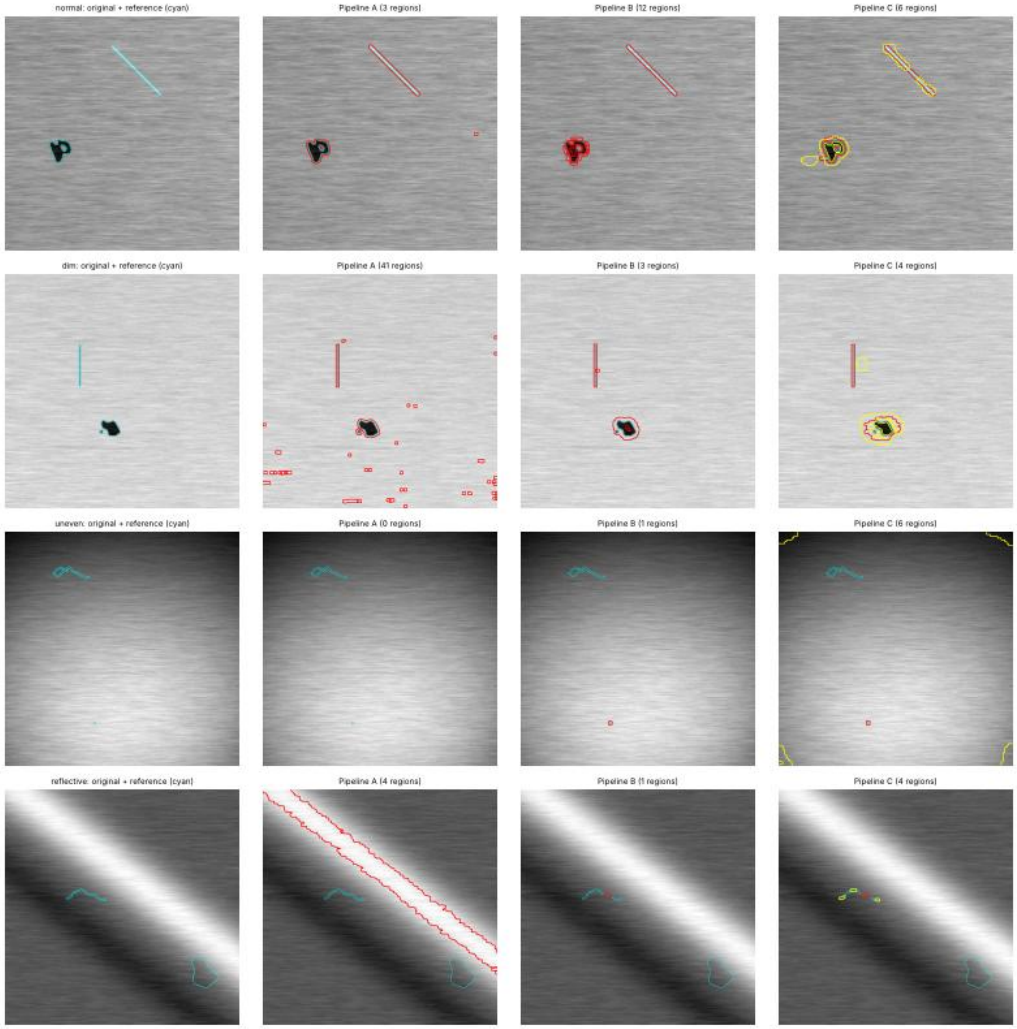


Figure 10.2 - Annotated outputs for four sheet conditions (rows): original with reference (cyan); pipelines A, B and C (red = reject region, yellow = uncertain region).

Results in words. Pipeline A has the lowest recall (28 %, only 9 % on reflective sheets) because the single global threshold is set by the global noise sigma, which includes the illumination gradient; on dim sheets it produces 23 false alarms per image and on unevenly lit sheets it finds nothing. Pipeline B is the most precise (89 %, 0.25 false alarms/image) but misses 53 of 107 defects (recall 51 %); its misses are mainly cracks (12), pits (14) and all 19 stains. Pipeline C has the best recall (60 %), IoU (0.211) and Dice (0.348), the best low-contrast recall (34 % against 24 % for B and 20 % for A) and finds 9 of 19 large stains through its large-area channel; its price is more false alarms (2.0 per image, 6.3 per image on unevenly lit sheets), and 179 of its 221 regions are marked uncertain for operator review rather than rejected automatically.

10.3 Ablation: removing one stage of pipeline C at a time

Variant	Recall	Precision	IoU	FA / img	Low-contrast recall	Missed (by type)	Time (ms)
Full pipeline C	60%	46%	0.211	2.00	34%	crack 9, pit 13, stain 9, scratch 11, pinhole 1	29.4

Variant	Recall	Precision	IoU	FA / img	Low-contrast recall	Missed (by type)	Time (ms)
- illumination flattening	53%	49%	0.043	1.48	29%	crack 9, scratch 10, pit 20, stain 8, pinhole 2, scale 1	19.2
- CLAHE (dim sheets)	63%	46%	0.228	2.13	41%	crack 7, pit 13, stain 9, scratch 10, pinhole 1	30.8
- denoise / smoothing	59%	42%	0.218	2.13	29%	crack 5, pit 14, stain 9, pinhole 2, scratch 14	33.7
- line-response stage	54%	42%	0.210	1.97	34%	crack 15, pit 13, stain 9, scratch 11, pinhole 1	16.4
- boundary linking	61%	49%	0.212	2.05	34%	crack 9, pit 13, stain 9, scratch 10, pinhole 1	29.6
- region validation rules	64%	43%	0.218	2.37	34%	crack 7, pit 13, stain 9, scratch 8, pinhole 1	28.1
- large-area (stain) channel	53%	59%	0.202	0.88	24%	crack 9, pit 13, stain 16, scratch 11, pinhole 1	25.5

Table 10.3 - Ablation on the 60 images (107 defects).

- **Illumination flattening** is the most important stage for segmentation quality: without it IoU falls from 0.211 to 0.043 and pit misses rise from 13 to 20 because the gradient of an uneven sheet is thresholded as if it were a defect.
- **Line-response stage:** removing it increases missed cracks from 9 to 15 and reduces recall on reflective sheets from 32 % to 24 %: hairline cracks are too thin for the blob score.
- **Large-area (stain) channel:** removing it increases missed stains from 9 to 16 but reduces false alarms from 2.0 to 0.9 per image - the cost of finding stains is about one extra uncertain region per image.
- **Denoising/smoothing:** without it false alarms on reflective sheets rise from 0.53 to 1.2 per image and 5 more scratches are missed (14 against 11 - the noise inflates the robust sigma and raises the effective threshold).
- **Boundary linking:** its removal slightly raises false-alarm count (2.05 vs 2.00) and reduces precision (0.49 vs 0.46) while leaving recall unchanged - a small effect on this data because only 6 cracks are fragmented.
- **Region validation:** without it all 158 regions are automatic rejects instead of 42 reject + 179 uncertain, false alarms increase from 2.00 to 2.37 per image and recall rises to 64.5 % (the rules remove a few true weak regions).
- **CLAHE for dim sheets:** removing it **improves** recall (62.6 % against 59.8 %) and low-contrast recall (41 % against 34 %) - CLAHE helps the eye but not the detector (as found in Task 2). It is therefore moved to the display path in the final recommendation.

10.4 Block-level decision table

Method	Evidence	Advantage	Limitation	What if not
Piecewise point transform (display)	T1: low-contrast contrast 6.5 -> 28.6 grey levels, CNR unchanged	Faint defects visible to the operator; 0.09 ms	Does not improve detectability; thresholds must be in the mapped domain	Faint scratches stay at 6-12 levels on the screen and are missed by the human reviewer
CLAHE 1.5 / 16 x 16 (display only)	T2: contrast x1.6, noise x1.6, 0.17 FA/img vs HE 146; T10 ablation: not helpful for detection	Even display on dim or unevenly lit sheets	Reduces CNR slightly; can create texture	Dim sheets look flat; with HE/AHE the display shows false features
Gaussian 3 x 3 smoothing (median 3 x 3 if impulse noise)	T3: 35.7 dB, 86 % scratch contrast kept; median +10 dB for S&P	Small noise gain, scratch preserved	5 x 5 or larger loses 25-40 % of scratch contrast; median deletes 0.5 mm pits	False alarms on reflective sheets x2.3 (ablation)
Flatten + Gaussian band-pass	T5: CNR 5.81 vs 4.20; no ringing; T10: IoU 0.043 without flattening	Removes illumination and highlights; fixed 10-13 ms	FFT border wrap; cut-offs depend on resolution	IoU falls 5x; uneven sheets generate false regions

Method	Evidence	Advantage	Limitation	What if not
Point + multi-orientation masks	T6: point recall 62 % / 68 % lines; max of four masks with half-response gap	Finds isolated and linear defects; fixed 3 x 3 cost	Orientation gap at 15-30 deg; texture response; low-contrast limit	Hairline cracks missed: 6 more cracks (T10 ablation)
Canny edges (boundary and review)	T7: F1 0.79, 0 fragments per crack, 0.003 % grain edges	Thin, continuous boundaries for measurement	Weak-edge recall 48 %; responds to sheet boundary	Boundaries from Sobel/Prewitt are thick, fragmented (1.75 per crack)
Local hysteresis threshold	T8: adaptive IoU 0.25-0.33 vs fixed/Otsu 0.003-0.005	Follows illumination; keeps weak parts attached to strong cores	Under-segments low-contrast defects	Fixed/Otsu flag 52 % of a good sheet
Connected regions + linking <= 5 px	T9: linking 7 px merges close pits; 5 px does not; fragments 7 -> 4	Closes crack gaps up to 0.6 mm; measures in mm	Cannot split touching defects; weak cracks missed	A broken crack is reported as several small defects
Region validation (area, rolling-mark, saturation rules)	T10: FA/img 2.37 -> 2.00; 158 automatic rejects -> 42	Controls false rejects; routes doubtful regions to review	Rules tuned on simulation; may drop weak real defects	Every flagged region would auto-reject (158 rejects instead of 42)
Large-area stain channel	T10: stain misses 16 -> 9	Finds broad low-contrast stains	+1.1 uncertain regions per image	Stains pass unnoticed

Decision rule for uncertain masks and defects near the acceptance limit (used by pipeline C): a region with score >= 1.6 and area >= 10 px (0.24 mm2) is an automatic **reject candidate**; a region with a smaller score or area, a region inside a saturated reflection, or a region whose measured area is within +/-30 % of the acceptance limit is **uncertain** and the sheet is paused for operator confirmation (the line may pause); a sheet with no region passes. Because 81 % of C's regions are uncertain, the first operating mode is "assist": no sheet is rejected automatically until real-line statistics show the uncertain-region rate and the operator agreement rate are acceptable.

10.5 Processing-time budget

Stage of pipeline C	Normal sheet (ms)	Dim sheet (ms)	Reflective sheet (ms)
1 condition analysis	0.2	0.2	0.2
2 CLAHE (dim sheets only)	0.0	1.1	0.0
3 flatten + denoise + blob score	11.4	11.4	12.8
4 multi-orientation line response	12.1	11.6	10.5
5 hysteresis threshold	0.7	0.5	0.5
6 boundary linking (dilate+label)	0.4	0.4	0.4
7 region validation + decision	4.3	1.9	1.9
8 large-area (stain) channel	1.9	1.5	1.2
Total per 512 x 512 window	**31.0**	**28.6**	**27.5**

Table 10.4 - Measured stage times (median of 12 runs, one core). Dominant stages: flatten + blob score (about 12-14 ms) and the line response (11-15 ms).

Check against 250 ms: one 512 x 512 window (80 x 80 mm) takes about 29-37 ms (total 28.7 ms on the common test image, 37 ms measured on a normal sheet), well inside the 250 ms target. The **maximum 600 x 400 mm region is about 38 windows at this resolution (3846 x 2564 px)**, i.e. about 1.1-1.4 s on one core. Spread over the 4 cores of the edge PC by tiling (ideal speed-up 4, a realistic 3), this gives about 0.4-0.5 s, which **does not meet 250 ms for the whole 600 x 400 mm region** when it is processed in full-resolution. The line may pause, but at 0.6 m/s the sheet covers 150 mm in 250 ms, so two simplifications are suggested. (1) **Drop the line-response stage on the first pass** and run it only on candidate regions: the ablation shows this reduces time to 19 ms (-34 %) and costs 6 points of recall mainly on cracks (these ROIs would then be re-checked with the line stage). (2) **Run the flatten and blob stages at half resolution** (0.31 mm/px) for the first pass, which cuts the cost of those stages by about 4 times, then re-examine only the flagged regions at full resolution; the 0.3 mm scratch is 1 px wide at half resolution so a half-resolution pass can only be a

pre-screen. These two simplifications were estimated from the measured stage costs, not measured end-to-end, and the simplified pipeline must be re-validated on the labelled set.

10.6 Final recommendation

Recommended pipeline: C (adaptive multi-stage), with CLAHE moved from the detection path to the operator display path, and run in "assist" mode with all regions reviewed until real-line data are available. Evidence: highest recall (60 %; 63 % without CLAHE), IoU (0.21 / 0.23), Dice and low-contrast recall of the three pipelines; handles all four illumination conditions (recall 78 / 83 / 59 / 32 % on normal / dim / uneven / reflective); every stage contributes in the ablation except CLAHE; 29-37 ms per 80 x 80 mm window.

An operating condition under which it is not reliable: (a) **strong illumination gradients** (the uneven sheets: 6.3 false alarms per image) where the background estimate (Gaussian sigma 25 px = 3.9 mm) cannot follow a gradient that changes faster than that; and (b) **large saturated reflections** (recall only 32 % on reflective sheets, because defects inside or on the edge of a saturated area are not visible in the data at all). In both cases the system flags the sheet as "illumination fault - re-capture" rather than passing it.

Validation test: (1) collect at least 300 real images over several shifts, including at least 30 of each defect type and 50 low-contrast cases, labelled by two inspectors (inter-inspector agreement recorded); (2) hold out half for validation, never used to set thresholds; (3) compute defect-level recall and false alarms per image with 95 % confidence intervals, separately for each material (stainless / mild steel) and condition; (4) run an A/B on the line comparing the system with the human inspector for at least two weeks; (5) acceptance targets agreed with the factory (for example recall >= 90 % for defects >= 0.5 mm, false alarms <= 1 per 10 sheets) before automatic rejection is enabled; (6) repeat the timing test on the edge PC with all 4 cores.

Periodic review for new defect types: (i) each week the images flagged as uncertain, and a random 1 % of passed sheets, are reviewed by an inspector; (ii) any defect found by humans but missed by the system, or any new pattern, is added to the labelled set with its type; (iii) each month the recall/precision are recomputed and compared with the validation baseline, and thresholds are re-calibrated on defect-free sheets of each batch; (iv) a drift check on the grey-step reference target at each shift; (v) the image set, masks and configuration are archived for 30 days as required (original images and result masks, with the configuration hash).

Limitation: all numbers come from simulated images; the pipeline parameters (hysteresis 1.15 / 0.9, rules in the decision table) have not been validated on real sheets, and recall on low-contrast defects (34 %) is far below any plausible acceptance target - this is a property of the simulation's 6-12 level defects against a 4-level texture, and shows that very faint defects need better optics/lighting (for example darkfield, raking light or polarisation), not only more processing.

Outcome vs prediction: Confirmed: A fails across conditions; B is precise but misses faint defects; C has the best recall/IoU and flattening is its most important stage. Not confirmed: I expected the stages up to the hysteresis threshold to be enough to give high precision - C produces 2.0 false alarms per image and needs the uncertain-region route; and CLAHE did not help the detection.